

Operation Manual

9375BSC Multi Card



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Chapter 1 Introduction

1.1 Preface

This manual only applies to 9375 BSC Multi Card machine. Before using the machine, please read the instruction manual carefully first. Please keep the manual properly so that consulting it at any time in the future.

Any question while operating please contact our technicians.

1.2 Specification

Items	Parameter	
Dimensions	Length*Width*Height (mm)	2815*706*1325
Power Source	AC 110V	
Power	900W	
Weight	220kg	
Infeed Paper Spec	Max (W*L)	370mm*650mm Max. length can be up to 900mm.
	Min (W*L)	210mm*210mm
	Gram Weight/Thickness	150g-400g/0.15-0.4mm
Output Size	Max (W*L)	370mm*900mm
	Min (W*L)	50mm*48mm
Cross Cutter	First Cut	3.2-15mm, It will chop up if it's larger than 15mm.
	Last Cut	5-15mm
	Gutter Cut	3-15mm, It will chop up if it's larger than 15mm.
	Max. cutting Qty per Sheet	32
	Accuracy for Cutting	±0.3mm
Slitter	Quantity	6
	Side Trim	3.2-38.5mm
	Gutter Slit	0 or 5-15mm
	Accuracy for Slitting	±0.3mm
Crease	Max Qty for Crease per Sheet	32
	Crease Direction	Both
	Standard crease Die Spec.	1.0mm
	Accuracy for Creasing	±0.2mm
Option tools	Scoring Tools	Option
	Cross Perforator Die	Options: 12/24TPI
	Rotary Perforator Assembly	Options: 12/24TPI
	Max Qty for Rotary Perf. Assembly	8 sets
Option tools	Strike Perf. Assembly and Max Qty	Option 12TPI,8 sets
Feeding Type	Top Vacuum	

Speed	A4 trimming 4 sides with on crease in middle	24 Sheets/min
Loading capacity	150mm	
Double Feed Detect	Available	
Skew Adjustment	Automatically	
Image Compensation	X/Y Direction Compensation	
User Templates Storage	60	

Notes: The machine is keep upgrading, specifications in this manual may be updated as per the change without being informed.

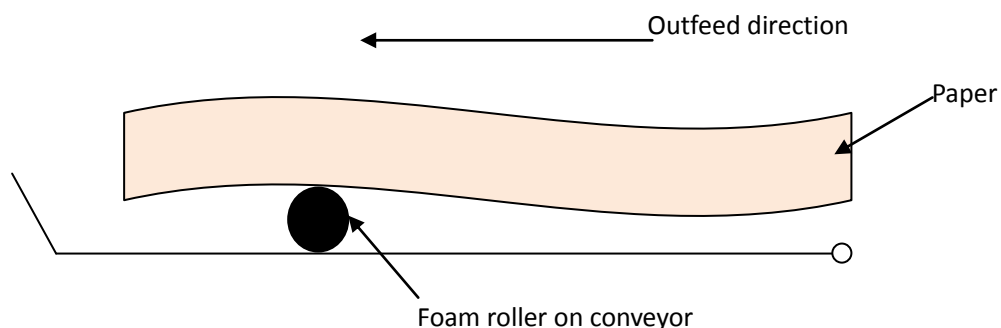
Important tips:

1.Double feeding:

- 1) If the two sheets of paper overlapped or distance between the leads of two double fed paper less than or equal to 200mm, the machine will automatically make them fall to the bottom tray and keep running until it has finished the setting batch or paper has run out. And then show **Double Feed Error**.
- 2) If the distance between the leads of two double fed paper larger than 210mm, the machine will stop immediately and give out **Double Feed Error**.

2. Auto conveyor:

- 1) If we didn't input any cutting data, the belt on the conveyor will not work and the pressure wheel on the conveyor will stop at the point where the operator had set.
- 2) The control button works only when the machine is on normal standby status.
- 3) When the finished paper is larger than or equal to 650mm, we need to manually put one sheet on the conveyor beyond the foam roller first. And remember to take the finished paper away from the conveyor in time. As shown in the following figure, the manually set paper is used to guide the coming finished long sheets so that it will not get stuck at the bottom of foam roller. That is to say the foam roller cannot be used for long sheets.



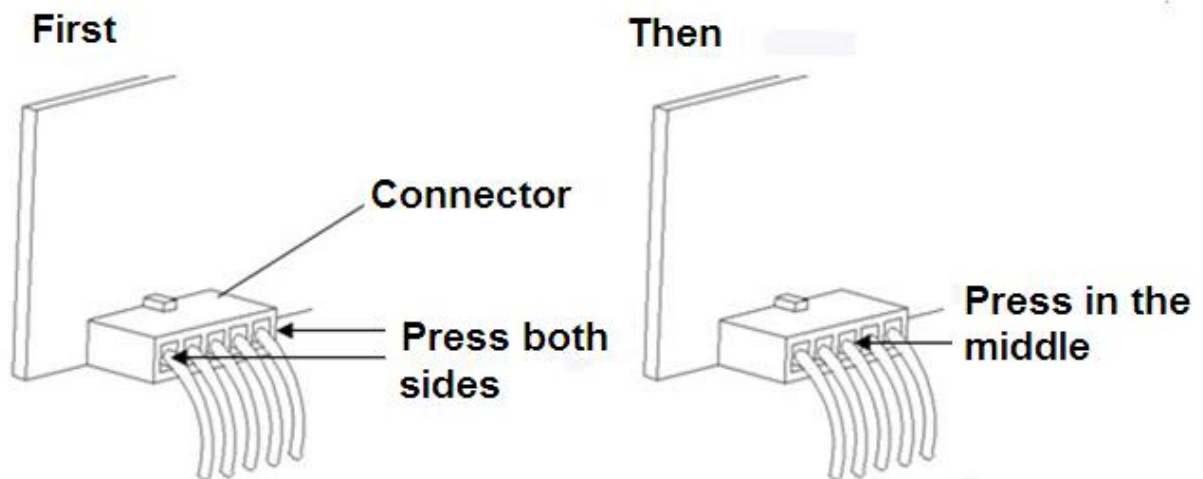
Two crease stations:

1. When we use the two crease bar-tools at the same time, the first crease line made by crease position1 must larger than or equal to 50mm.

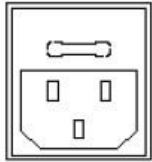
Chapter 2 Safety

Please shut down the machine before the maintenance. In order avoid any mistake when re-connecting wires back, we advise you to mark each connector before replace of any electrical component.

Mind that every connection should be firm, first hold both sides of the connector, and then, in the middle, not firm connection is possible to damage the electrical components.

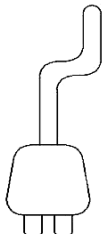


Power supply



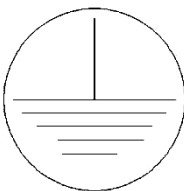
AC input, Voltage/Frequency is:
AC 110V 50Hz/60Hz.

! If the voltage or frequency are not in the range, the machine may not be used normally or even may cause unrecoverable damage



Power cord, Over 10A current is endurable

! 1. Do not remove the cord when the machine is running, illegal operation may cause component damage or machinery jam
2. Remove the cord when you are installing components.



Ground symbol. If you see this symbol in any place, please do not touch it anyhow, or uninstall it.

! If the ground is not well connected, it may cause human body injury.

2.1 Environment

Temperature: 18°C to 28°C

Humidity: 40% to 70%

Altitude: Below elevation 1000m

There is no corrosiveness gas, flammable gas, oil mist and so on in room

2.2 Do's and Don'ts



Do-Read this manual and fully understand before the operation.



Do-Check the *plug* and *machine voltage and frequency* to your main supply, and that the socket has a correct working earth lead for this single insulated machine



Do make sure all safety covers are in place. The top covers have an interlock switch which will disable the unit if removed



Do open or close the cover slowly



Do contact the local maintenance center before you are about to move the machine.



Do disconnect the power before clean the inner side.



Do unplug the cord if you won't use the machine for a long while,



Do be careful of the blade edge



Don't install the machine on an unstable ground.



Don't operating with wet hand, especially plug or unplug the cord.



Don't wear long hair, loose fitting clothes or put your fingers into the creasing

unit nip, while the operation..



Don't place any receptacles with any liquid on any surface of machine.



Don't put other pieces, especially tiny pieces on loading table.



Don't alter or uninstall the machine, unless by *Dumor* authorized engineer.



Don't touch any running parts while running



Don't shut down the machine while running.



Don't put heavy matter on machine or shock it.



Don't put the side guide under the feeding system.

2.3 Operating Instructions

English

Please read these operation instructions before putting the machine into operation and observe the safety precautions.

Model: 9375BSC

Input: 110V

1. Children must not operate the machine!
 2. Do not reach into the machine!
 3. Long hair can become entangled in cutting head!
 4. Take care of ties and other loose pieces of clothing!
 5. Remove any paper clips!
 6. Components which may endanger the operator are covered!
 7. Connect the power cord to a single phase socket 250 V / 10 A.
 8. Ensure free access to mains.
 9. Disconnect from the mains if not used for a long period.
 10. Top fold plate must lock into position correctly.
 11. Remove the plug before cleaning the machine! Do not perform any repairs on the paper folding machine!
 12. the socket-outlet shall be installed near the equipment and shall be easily accessible.
- this equipment is not intended for use by children (the product is not a toy);

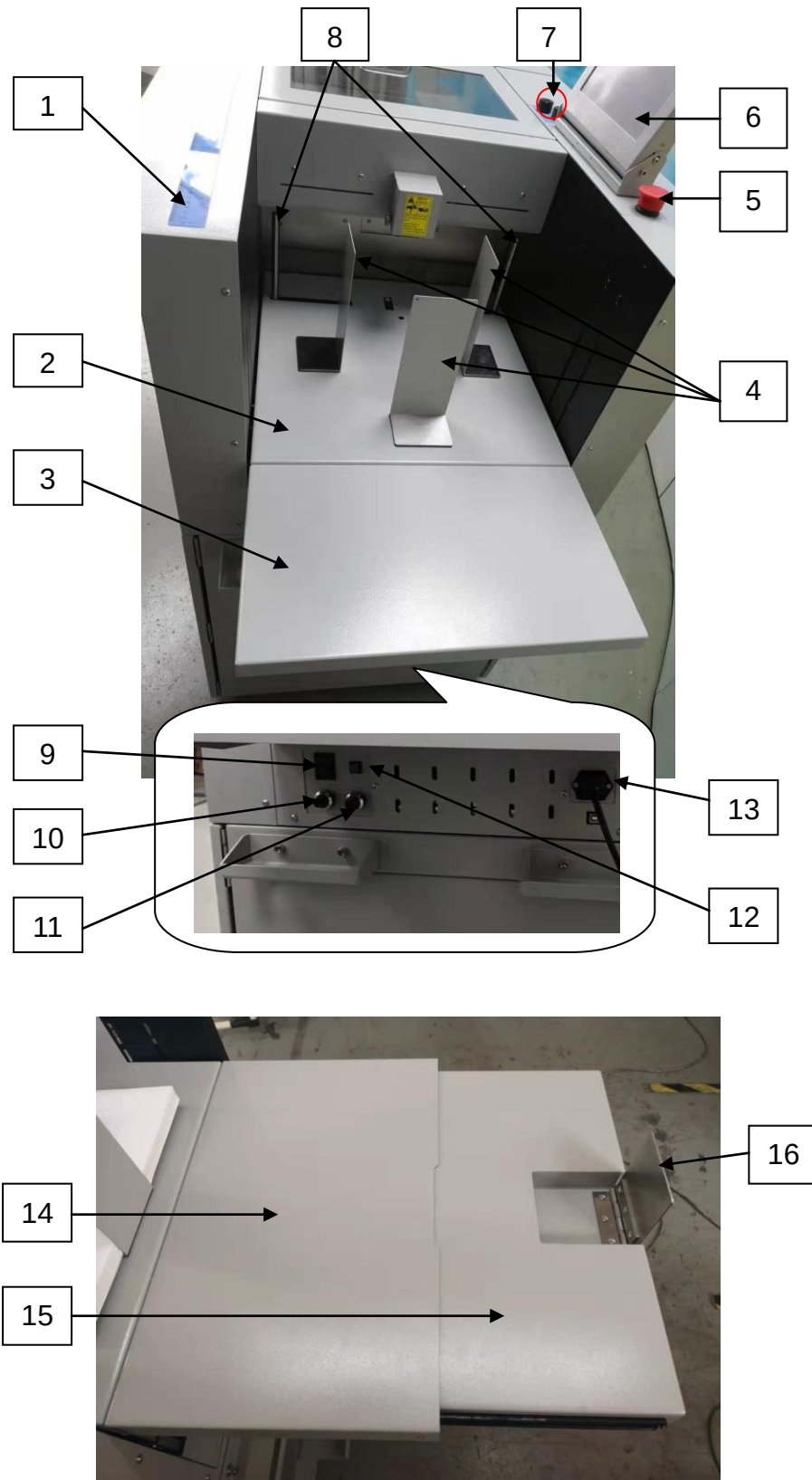
-
- avoid touching the document/media feed opening with the hands;
 - avoid clothing touching the document/media feed opening;
 - avoid hair touching the document/media feed opening; and
 - keep aerosol products away [for equipment incorporating a universal (brush) motor only.

Be careful of any metal or flammable thing in internal machine, or it may cause fire or electronic shock. If it happens, first shut down the power, disconnect the cord, and then contact the technician.

If machine becomes heat, smoke, or smelly, shutdown at once, disconnecting the cord, and contact the maintenance staff.

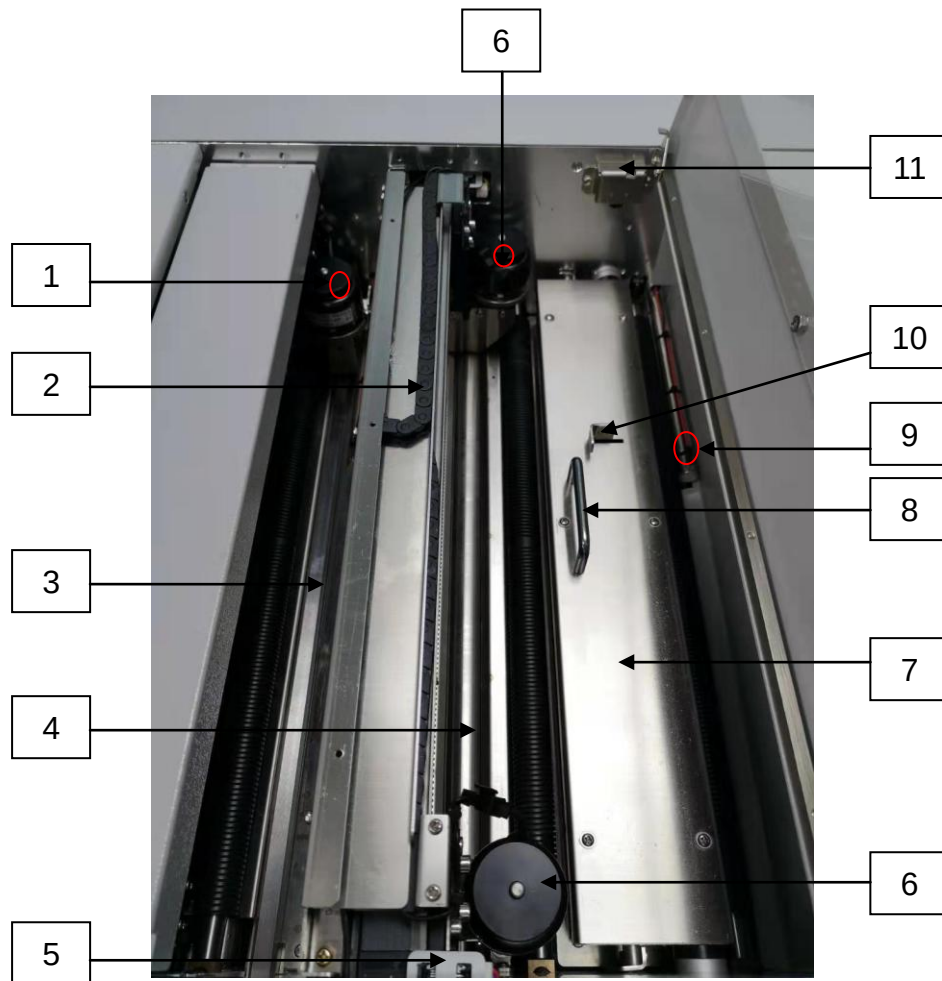
Chapter 3 Main Parts and Assy

3.1 Feeding unit



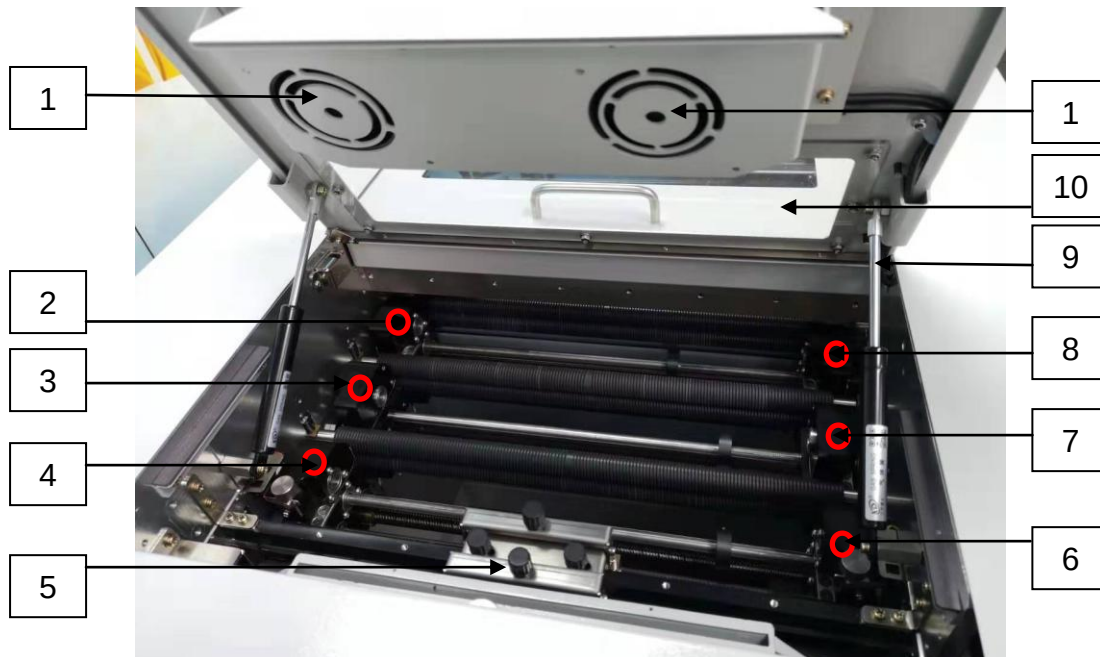
No.	Description	Remarks
1	Ruler bar	It's used to measure the length of sheets.
2	Feeding table	It's used to put the sheets on.
3	Feeding extension table	It's used to support the longer sheets.
4	Stop guide	It's used to make the sheets squared on the table.
5	Emergency switch	It's used for emergency stop.
6	Touch screen	Display
7	Blow and segregator adjust knob	It's used to adjust the blowing and the double feeding.
8	Side pillar	It's used to keep the sheets squared.
9	Power switch	Power on or off.
10	Suction adjust knob1	Suction power adjustment on auto skew adjuster.
11	Suction adjust knob2	Suction power adjustment on feeder.
12	Table release button	It works together with START or STOP button.
13	Power socket	Plug the power cord.
14	Feeding extension table1	For longer sheets. (Only the new version)
15	Feeding extension table2	For longer sheets. (Only the new version)
16	Back stop guide	Keep the tail of sheets. (Only the new version)

3.2 Perforate unit



No.	Description	Remarks
1	Cross partial perforator	It's used to do the cross partial perf line.
2	Wire protecting chain	It's used to protect the wires of perforator.
3	Perforating anvil	It's used for cross partial perforator.
4	Perforating anvil	It's used for linear partial perforator.
5	Home position sensor	It's used for the linear partial perforator.
6	Linear partial perforator	It's used to do the linear partial perf line.
7	Barcode reader box	The barcode reader is fixed inside.
8	Handle	It's used to turn the box over.
9	Infeed sensor	Feeding detect.
10	Paddle	Push to the right will unlock the box.
11	Cover secure switch	Top cover secure.
12	Handle for perf anvil	It's used to take out the cross partial perf anvil.
13	Side cover for perf anvil	Open the cover to check the anvil.
14	Handle for perf anvil	It's used to take out the linear partial perf anvil.

3.3 SC Unit



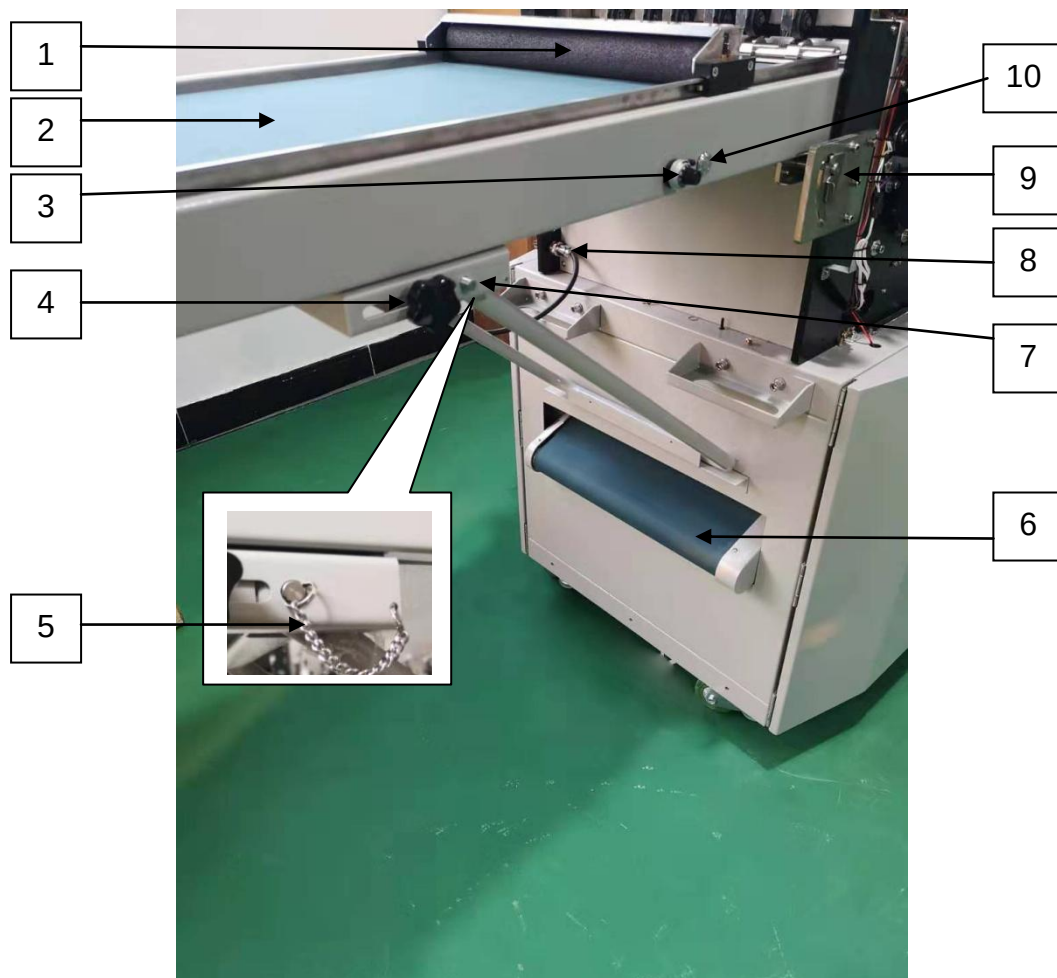
No.	Description	Remarks
1	Blowing fan	It's used to blow the cutter bleed down.
2	Slitter#1	Side trim for non-op side.
3	Slitter#2	Middle slitting.
4	Slitter#3	Gutter slitting
5	Lag kit wheel	It's used to add pressure on the rollers near the cutter. It's helpful for cutting stability.
6	Slitter#4	Gutter slitter.
7	Slitter#5	Middle slitting.
8	Slitter#6	Side trim for op side.
9	Gas spring	Support the top cover.
10	Window	It's used to help look through inside.

3.4 Crease Unit



No.	Description	Remarks
1	Top cover secure switch	For the top cover at crease unit.
2	Adjust shaft for crease bar #2	For crease bar #2.
3	Mounting bar	For exit rollers and optional tools.
4	Exit fixed shaft	Mount the anvil for optional tools.
5	Exit rollers	To drive the paper out.
6	Plug	For linear partial perforators
7	Depth adjust knob	It can be used to adjust the crease depth. Factory used only.
8	Adjust shaft for crease bar #1	For crease bar #1.

3.5 Conveyor



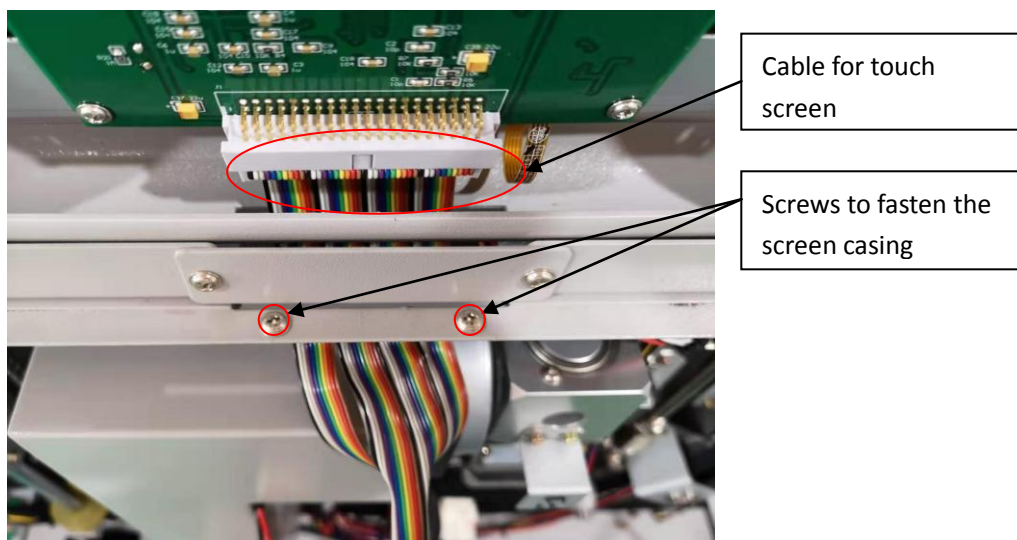
No.	Description	Remarks
1	Foam roller	It's used to drive and stack the cards.
2	Conveyor belt	Drive the cards.
3	Speed adjust knob	Adjust the speed of conveyor belt.
4	Fixed screw	Fasten the support shaft.
5	Fixed chain	Use to hang the fixed shaft.
6	Waste discharge belt	Use to drive the waste out.
7	Fixed shaft	Use to fix the support shaft.
8	Conveyor cable	For control.
9	Clip	Use to fasten the conveyor.
10	Discharge button	Use to collect all the cards on the conveyor.

Chapter 4 Installation

4.1 Touch Screen Installation

Remove the wooden case and take machine out. Find the touch screen from the waste box. Then install it as following pictures:

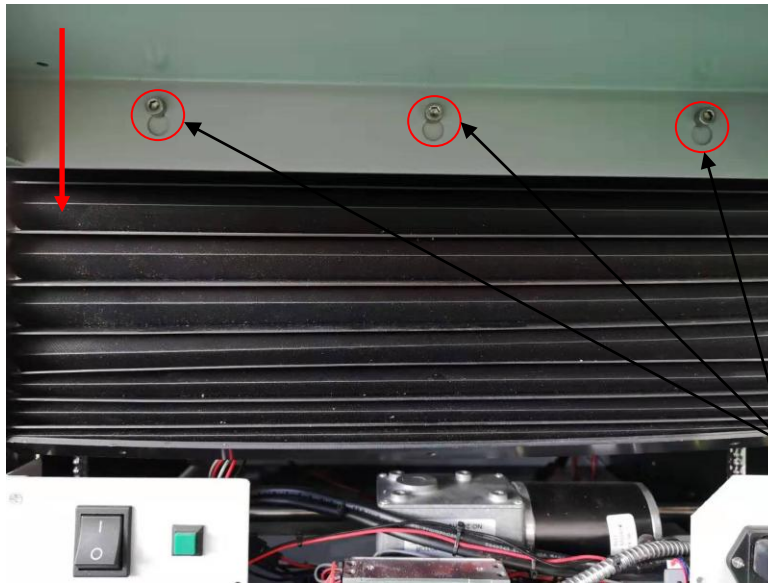
1. Plug the cable for the touch screen first, make sure the cable is connected well;
2. Fasten the screws for screen casing.



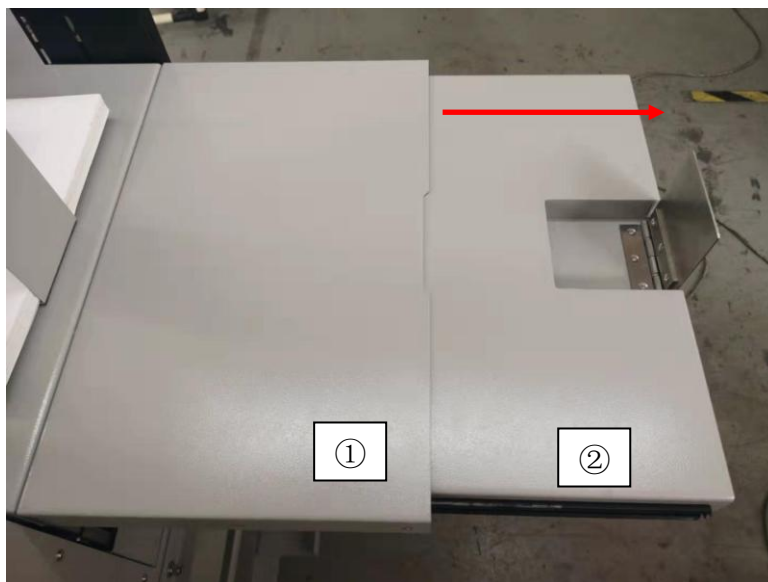
4.2 Feeding table installation

As shown in the following picture, Feeding extension table ① and ② are fixed together as one part. So we just need to hang this part to the feeding table by fastening the three hex screws. Shown in the picture.

And when we pull the extension table out, we need to pull it straight. Because it has guide rails on both sides. It may get stuck if the table was pulled askew.



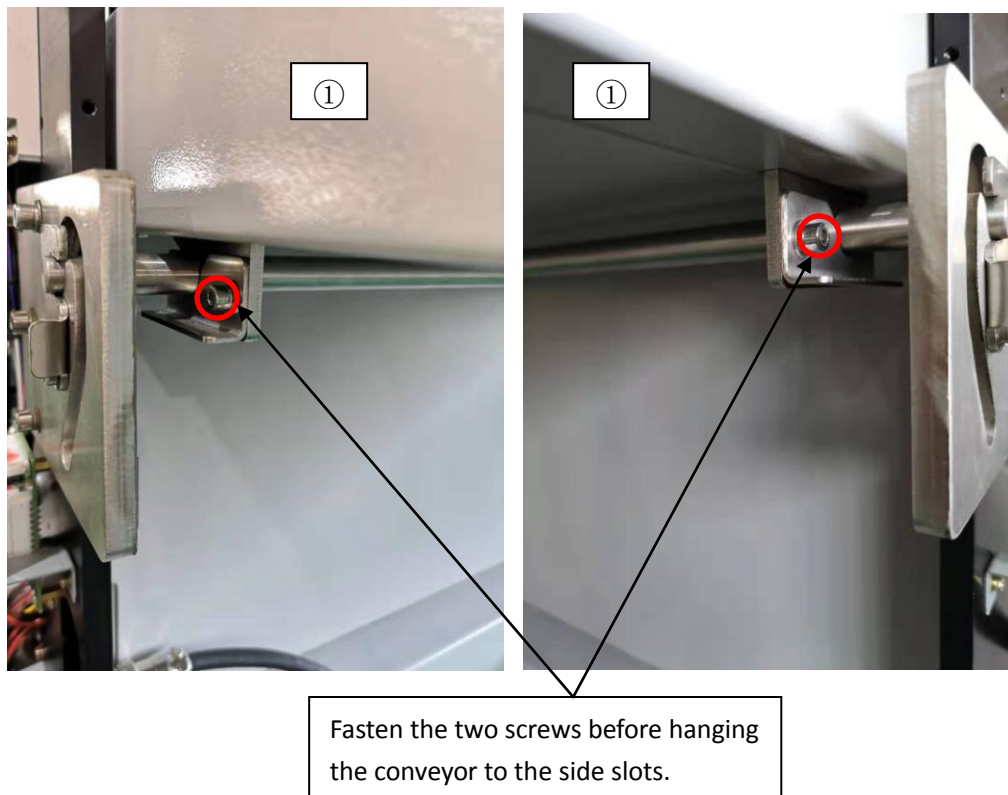
Drive the three screws out a little first, and then hang the extension table on by aligning the bigger hole. Then push the table extension down and fasten the screws.



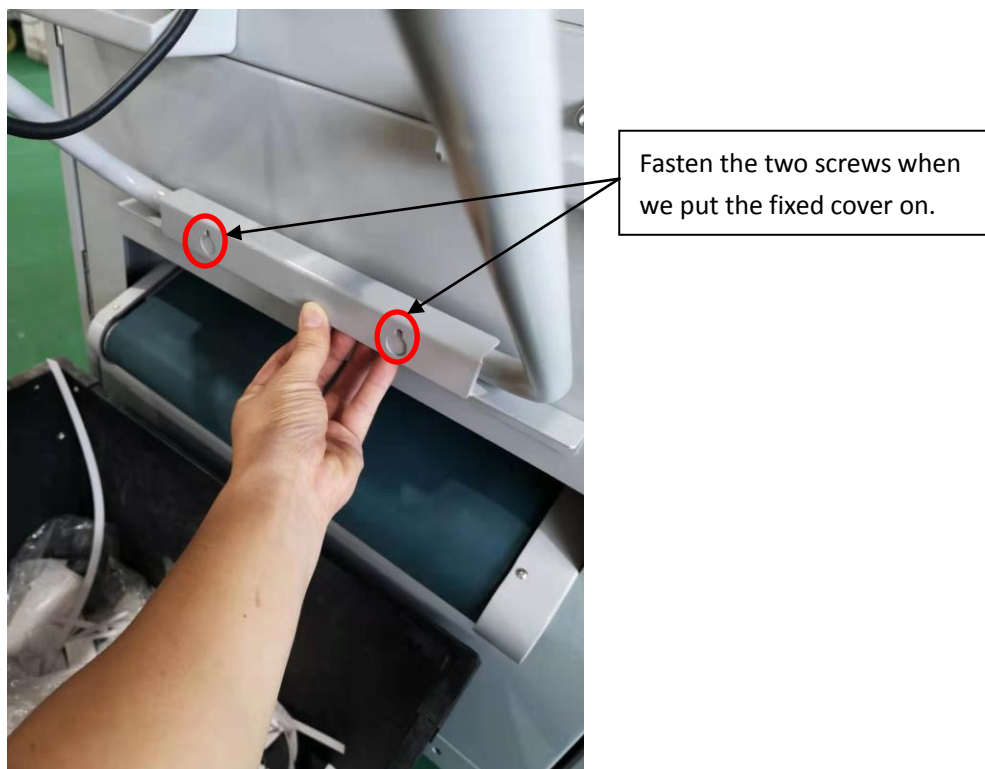
Drive the three screws out a little first, and then hang the extension table on by aligning the bigger hole. Then push the table extension down and fasten the screws.

4.3 Conveyor Installation

First, fasten the screws in picture ① before hanging the conveyor on the machine.



Then, put the support shaft to the bottom slot and lock it by the fixed cover. Shown in the following picture.



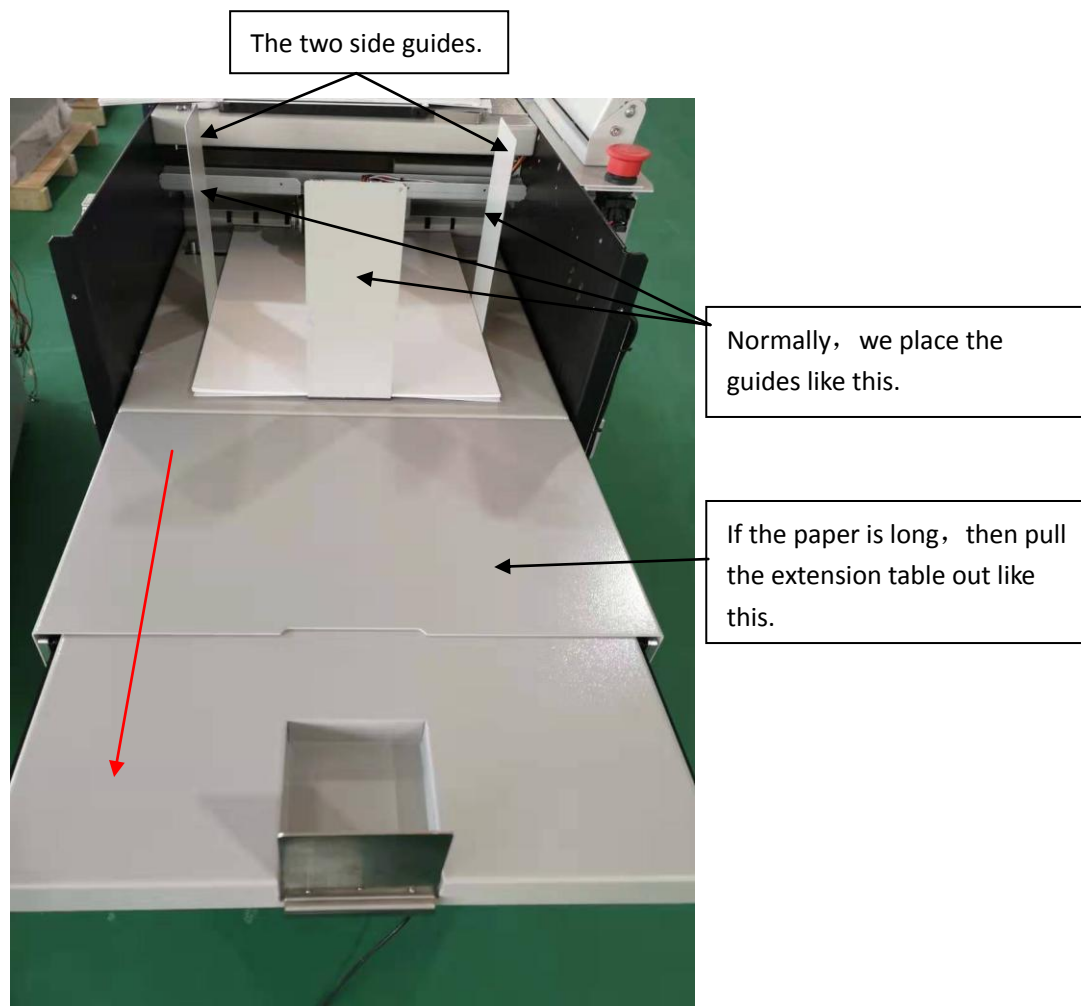
Finally, pull the clips to lock the shaft of conveyor by following the arrow to make it tight.



4.4 Deflectors and Side Guide Placement

Place the deflectors and side guide as following picture.

- The two side guides need to be put as front as possible. Because the paper lead is inside the machine a lot, if the two side guides is closed to the tail, the lead may not be fed stably enough.



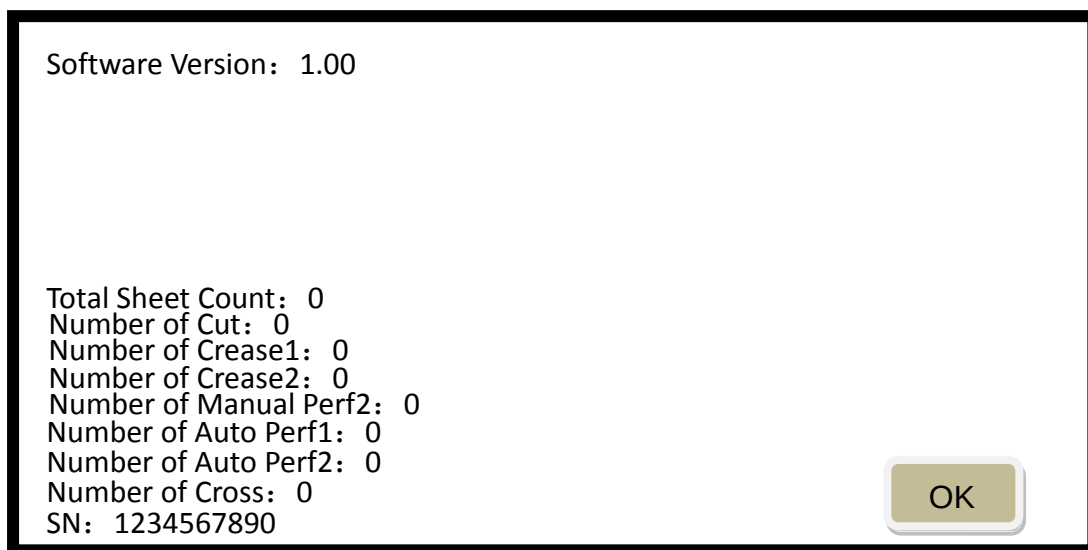
Chapter 5 Operation

5.1 Operation interface

In this section, we will introduce all the basic interfaces and the meaning of them.

5.1.1 Welcome screen

When we just power on the machine, it will enter to the welcome screen. As shown in the following, we can see the basic and historical information for the machine. The software version, serial number, the job count and so on.



5.1.2 Main screen

After we click **OK**, it will enter to the main screen. We can go the set up a job, run a job, check the existing jobs and so on.

Edit: It's used to set up a job manually.

Smart Input: It's used to set up a job in a much faster way.

Recall Job: It's use to check the user's saving jobs. Every time the users use Save as jobs, they will be saved under this option.

Template: It's used to save the factory recommending jobs. We can find and select a lot of common jobs under this option.

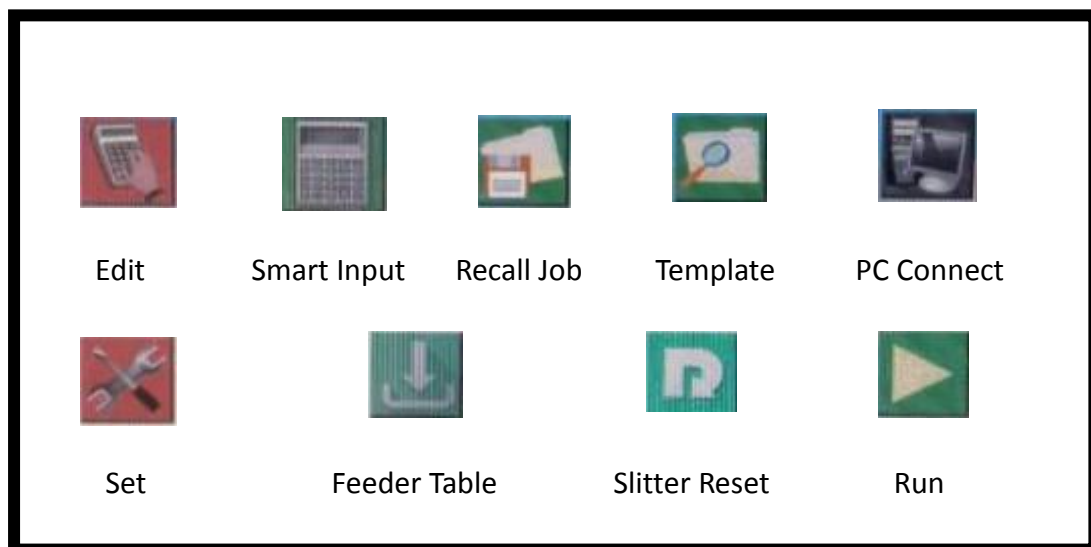
PC Connect: It's used to show whether the machine is connected to the computer or not. If the machine is connected to computer successfully, the icon will be lighted on. It's basically for Trimposer.

Set: We can enter the service mode and do some basic settings before we start to set up a new job.

Feeder Table: It's used to drive the feeder table to the bottom. And it can be used to fix the elevator error together with the release button when the feeder table stuck on the top sometimes.

Slitter Reset: It's used to drive the all the slitters back to home position.

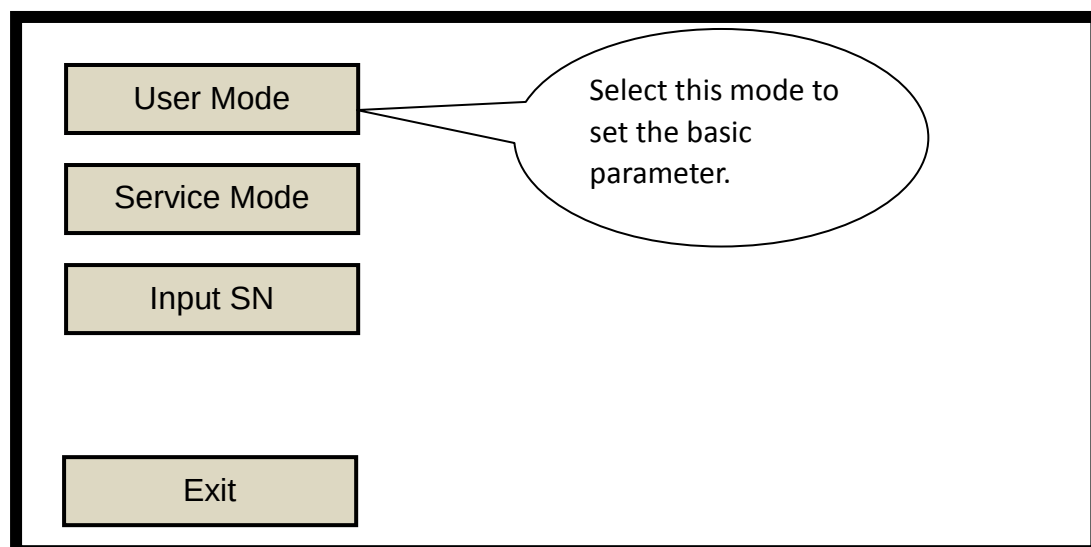
Run: It's used to enter the last running job.



5.2 How to start a new job

Before we start to set up a new job, we should enter to the user mode to set the basic parameters first.

Click the Set option and select the User Mode.



Then we will be able to enter to the following interface.

As shown in the following, we need to select the unit first. Adjust the brightness of touch screen according to the user.

mm/inch: The unit options. Click to select different unit.

Double feed detect: It's used to select the double feed detector **ON** or **OFF**.

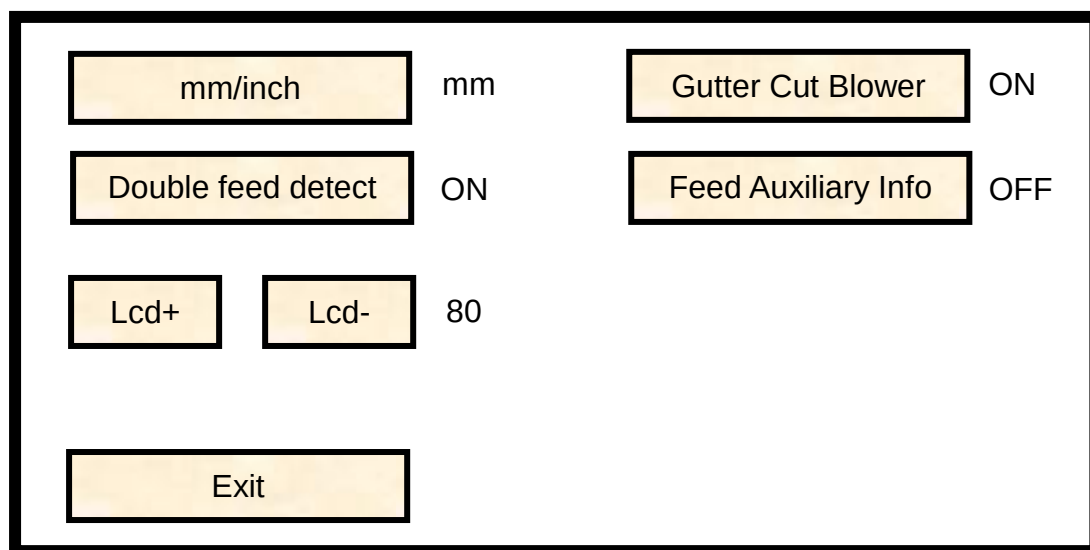
Sometimes, we can select to turn the detector off to run some sensitive sheets, especially when it only feed one sheet but keep showing double feed error.

Lcd+/Lcd-: It's used to adjust the brightness of touch screen. "Lcd+" means increasing the light while "Lcd-" means decrease the light.

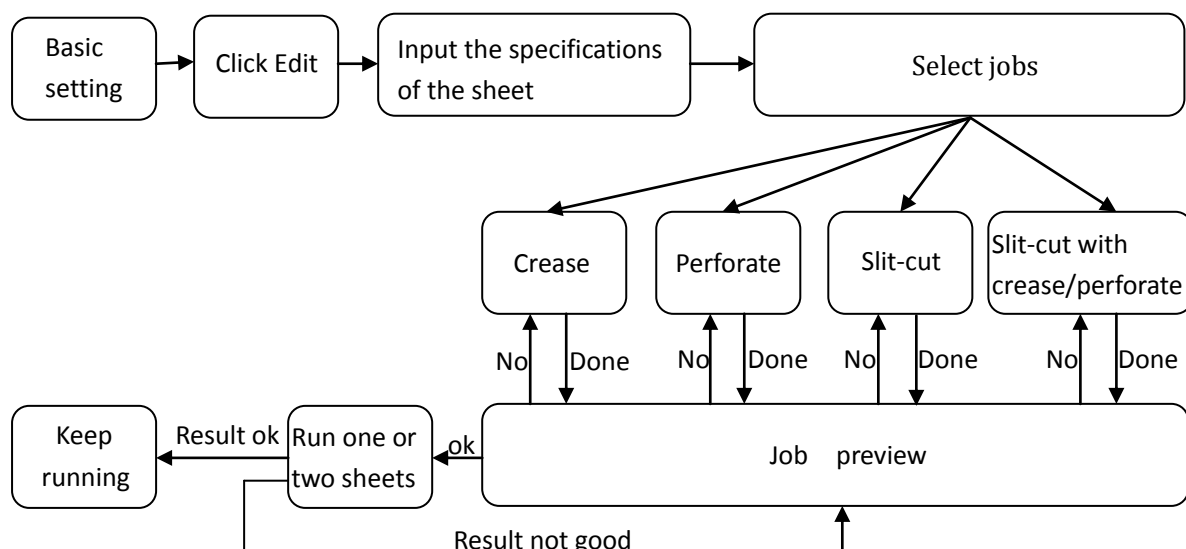
Gutter Cut Blower: It's used to turn on or off the blowing fans over the cutter. When we found the cutting waste is sticky and not going down to the waste bin, we can select to turn on the blowers to blowing it down. It's very useful when the weather is dry and the print full of statics.

Feed Auxiliary Info: It's used to give the operator a message that the machine has supporting parts for running the wider and thinner sheet. Because these parts need to be pushed out from the home position manually, it can use this option to show a message to the operator.

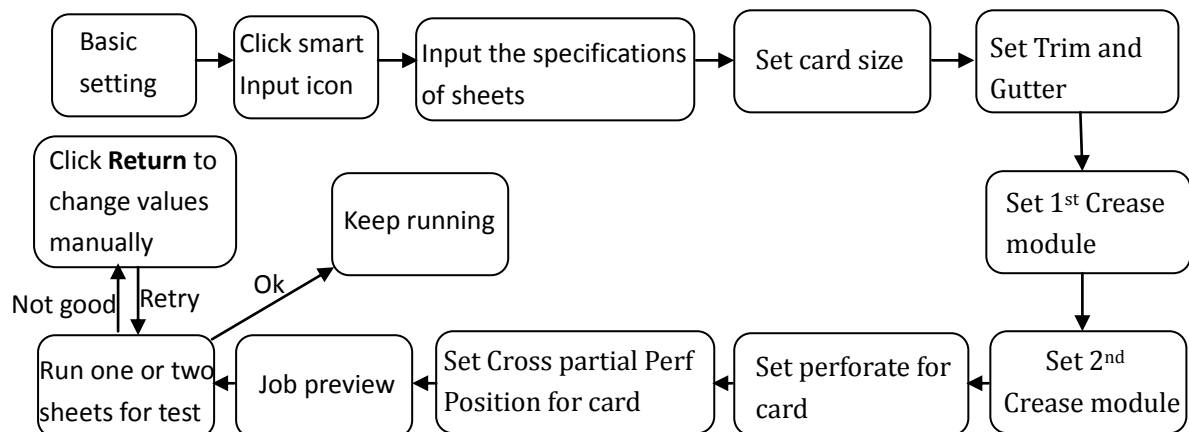
Exit: It's used to go back to the main screen after finishing setting the basic parameters.



1) Set up a job by manually input



2) Generate a job by smart input



5.2.1 Manually input

Click **Edit** to enter the manually input interface. As shown in the following.

The screenshot shows a software interface for manual input. At the top, there's a title bar with a red asterisk and the text "(Job name)". Below this, there are several icons: a house icon labeled "Home", a document icon with a plus sign, a scissors icon labeled "Slit", another scissors icon labeled "Cut", a ruler icon labeled "Bar-tool1", and a ruler icon labeled "Bar-tool2". The main area contains a table with 6 columns and 15 rows. The first column has values: 297.0, 210.0, 0.20, 0, 0, 2, 2, and then several empty rows. The other columns are labeled "Slit", "Cut", "Bar-tool1", and "Bar-tool2", and they all contain the value "0". To the right of the table, there's a section titled "Print drift compensation" with four rows, each containing a person icon and the value "0". At the bottom right, there's a green "Run" button. A "Page2" label and "(Next page)" text are also visible.



: Click this icon to enter the set paper size interface. We can change the length and thickness of sheets, position of register mark and so on under this interface . We

1.Set paper size

Length:

450.0

Width:

320.0

Thickness:

0.20

CR1 Depth:

2

Auto

CR2 Depth:

2

Auto

Reg slit mark:

0

Reg cut mark:

0

Back

Next

mm

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

SRA3

A3

330x480

320x460



: Click this icon to set the slitting values. There are 6 slitters. It doesn't mean we need to set a value for every slitter. It depends.

Set slit data

Home

Return

1

0

2

0

3

0

4

0

5

0

6

0

mm

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent



: Click this icon to set or change the cutting values. It can set 32 cuts in total. Click **Next** to enter the next page.

Set cut data

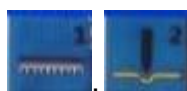
Home

Return

1	0	7	0	13	0
2	0	8	0	14	0
3	0	9	0	15	0
4	0	10	0	16	0
5	0	11	0		
6	0	12	0		

Next

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent



They are used to input or change the crease/perf values. The machine has two cross bar-tool stations-crease1 and crease2. When the operator insert the cross perf bar, the icon will show the comb. As shown in image 1.If the operator insert a cross crease bar, it will be shown like image2.

Set crease1 data

Home

Return

1	0	7	0	13	0
2	0	8	0	14	0
3	0	9	0	15	0
4	0	10	0	16	0
5	0	11	0		
6	0	12	0		

Next

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

Set crease2 data

Home
 Return

1	0	7	0	13	0
2	0	8	0	14	0
3	0	9	0	15	0
4	0	10	0	16	0
5	0	11	0		
6	0	12	0		

Next

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

Click **Page2** will enter the next page.

* (Job name)

Home
 Save As

	Slit	Cut	Bar-tool1	Bar-tool2
297.0	0	0	0	0
210.0	0	0	0	0
0.20	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
2	0	0	0	0
2	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

Page2

(Next page)

Print drift compensation

	0	
	0	
	0	
	0	

Run

As shown in the following picture. We can set the values of partial tools.

* (Job name)

Home

Save As

20.0	50.0	0	0	0
60.0	100.0	0	0	0
80.0	0	0	0	0
130.0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

Page1

(Next page)

Print drift compensation

0 0 0 0

Run



: It's for the partial perf tool1 in the exit unit. Click the icon to go to set the values. Shown in the following image.

Set linear partial perf1 data

Home Return

Position : 25.0

1	20	4	0	7	0
	30		0		0
2	0	5	0	8	0
	0		0		0
3	0	6	0		
	0		0		

Next

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

=AP1 =AP2

=MP1 =MP2

=AP1: It means the current input equals to the linear partial perf1 data. "AP" is the auto position.

=AP2: It means the current input equals to the linear partial perf2 data. "AP" is the auto position.

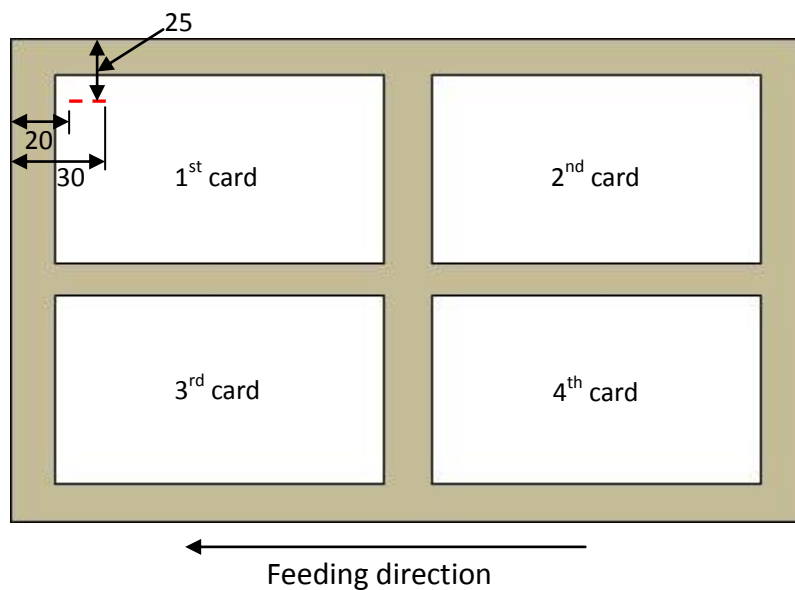
=MP1: It means the current input equals to the manually moved partial perf1

data."MP" is the manual position.

=**MP2**: It means the current input equals to the manually moved partial perf2 data."MP" is the manual position.

Position: It means the distance from the side edge of paper to the perf line. For example, we want to set a perf line on the first card. The position is 25mm away from the side edge. And the perf line starts from 20mm to 30mm. The length of the perf line is 10mm. The perf tool will be automatically driven to the position.

So as for a partial perf job at the feeding direction, we need to set the side position first and then set the start and the end of the perf line. Shown in the following picture.



: It's for the partial perf tool2 in the exit unit.

Set linear partial perf2 data

Home
 Return

Position : 150.

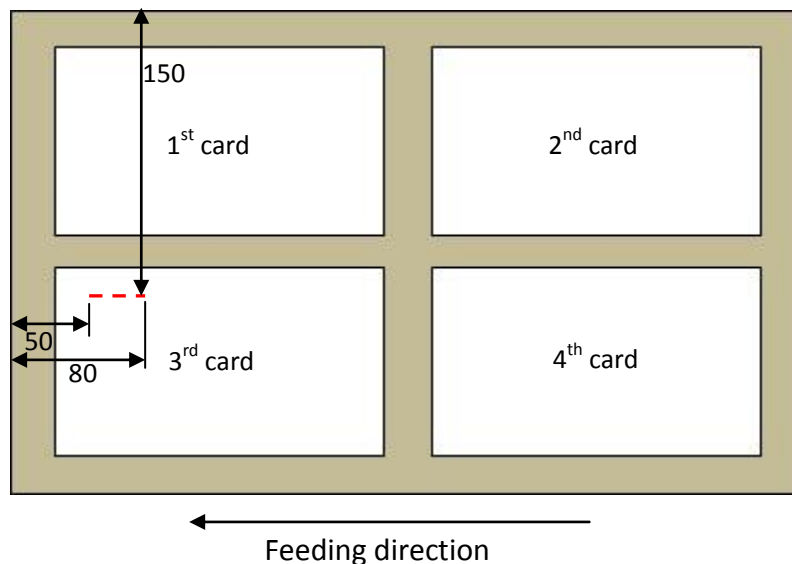
1 50 80
2 0 0
3 0 0

4 0 0
5 0 0
6 0 0

7 0 0
8 0 0
Next

1 2 3 -
4 5 6 +
7 8 9 Esc
. 0 C Ent

=AP1 =AP2
=MP1 =MP2



: It's for the manually moved horizontal partial perf tool1. As shown in the following picture, there is no position needing to be input. Because we need to move the tool manually to the position. But we need to input the start and the end of the perf line.

If we select “=AP1”, it will copy the data of linear partial perf1 to the perforate1. The same as the other features.

Set perforate1 data




Home

Return

1

0

0

2

0

0

3

0

0

4

0

0

5

0

0

6

0

0

7

0

0

8

0

0

Next

1

2

3

-

4

5

6

+

7

8

9

Esc

.

0

C

Ent

=AP1

=AP2

=MP1

=MP2



: It's for the manually moved horizontal partial perf tool2. It's in the same method to set the data as the perforate1.

Set perforate2 data




Home

Return

1

0

0

2

0

0

3

0

0

4

0

0

5

0

0

6

0

0

7

0

0

8

0

0

Next

1

2

3

-

4

5

6

+

7

8

9

Esc

.

0

C

Ent

=AP1

=AP2

=MP1

=MP2



: It's for the cross partial perf tool. First, we need to enter the position of the cross partial perf line. For example, 50mm for the first cross partial perf line and 100mm for the second cross partial perf line.

Set cross partial perf position

 Home
  Return

1	50.0	1	...	9	0	...
2	100.0	1	...	10	0	...
3	0		...	11	0	...
4	0		...	12	0	...
5	0		...	13	0	...
6	0		...	14	0	...
7	0		...	15	0	...
8	0		...	16	0	...

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

Then, set the first cross partial perf line from 60mm to 80mm.

Set cross partial perf data

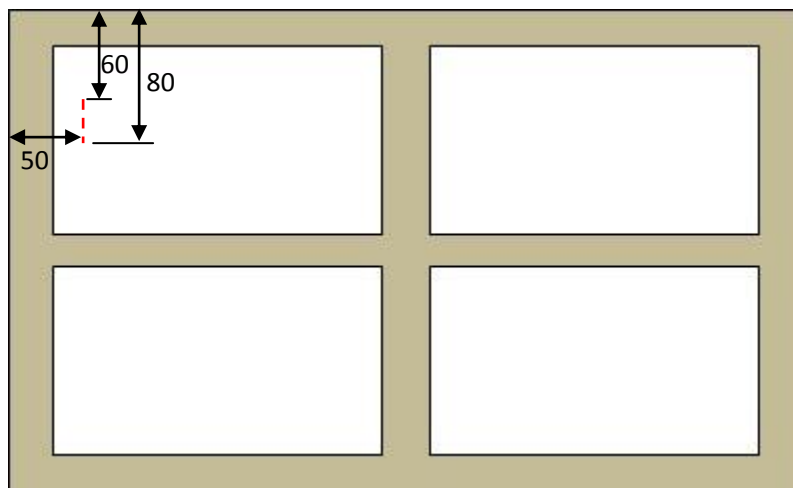
 Return

1	60	4	0	7	0
	80		0		0
2	0	5	0	8	0
	0		0		0
3	0	6	0		
	0		0		

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

=	C1
---	----

The result is as following:



In conclusion, we need to set the position first before we set the start and the end of the partial perforate line.

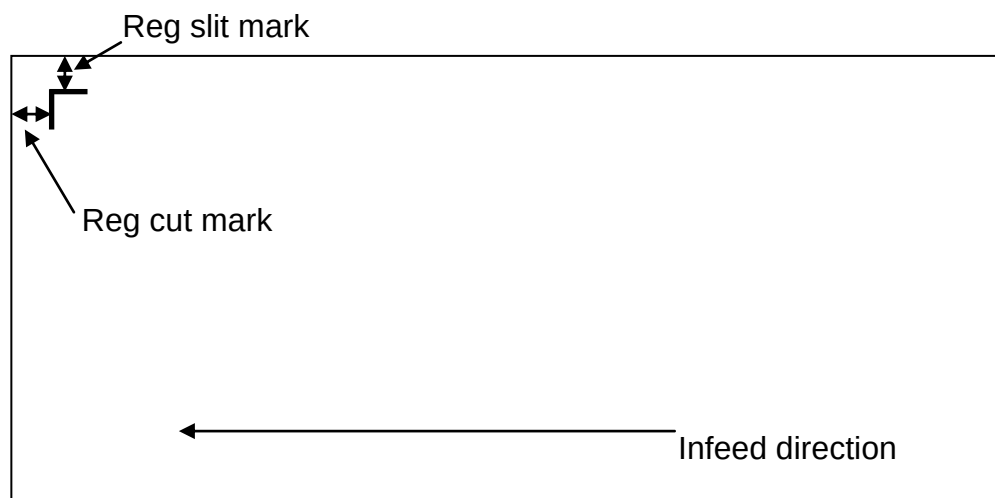
5.2.2 Smart input



1. Click smart input icon. It will enter the following interface. In the set paper size, we need to input the paper details. CR1 and CR2 depth are the crease depth. They depends on the thickness of the paper. After we input the thickness of paper, click auto. It will automatically show the depth recommendation for crease depth. It's empirical values. Select the recommended values can get a better result.

1.Set paper size				mm			
Length:	450.0			1	2	3	-
Width:	320.0			4	5	6	+
Thickness:	0.20			7	8	9	Esc
CR1 Depth:	2	Auto		.	0	C	Ent
CR2 Depth:	2	Auto					
Reg slit mark:	0						
Reg cut mark:	0	Back	Next				
				SRA3		A3	
				330x480		320x460	

As shown in the following picture, the Reg slit mark is the distance from the top side edge of paper to the top side edge of register mark. The Reg cut mark is the distance from the lead edge of paper to the lead edge of register mark.



- Next, it will need to input the card size. We have 50,54,55,210 and so on for recommendation.

The screenshot shows a software interface titled '2.Set card size'. On the left, there are input fields for 'Card Length: 54.0' and 'Card Width: 85.0'. Below these is a diagram of a yellow rectangle with 'Length' and 'Width' labels and arrows. At the bottom are 'Back' and 'Next' buttons. On the right is a numeric keypad with buttons for digits 1-9, 0, a decimal point, a minus sign, a plus sign, 'Esc', and 'Ent'. Below the keypad are four buttons labeled '50', '54', '55', and '210'.

- Then we need to set the Trim and Gutter. We also have recommending trim and gutter for a normal job.

***Please notice that the recommending values may not match the specific templates. Then we need to do some calculations to make sure the all the values are in our tolerance.**

3.Set Trim and Gutter

Lead Trim :
Side Trim :
Gutter Cut :
Gutter Slit :

Back

Return

mm

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

7

8

10

12

As shown in the following picture, it gives where the four parameters belong to.
Lead Trim: 3.2mm~15mm.It will be automatically chopped up if it was set over 15mm.

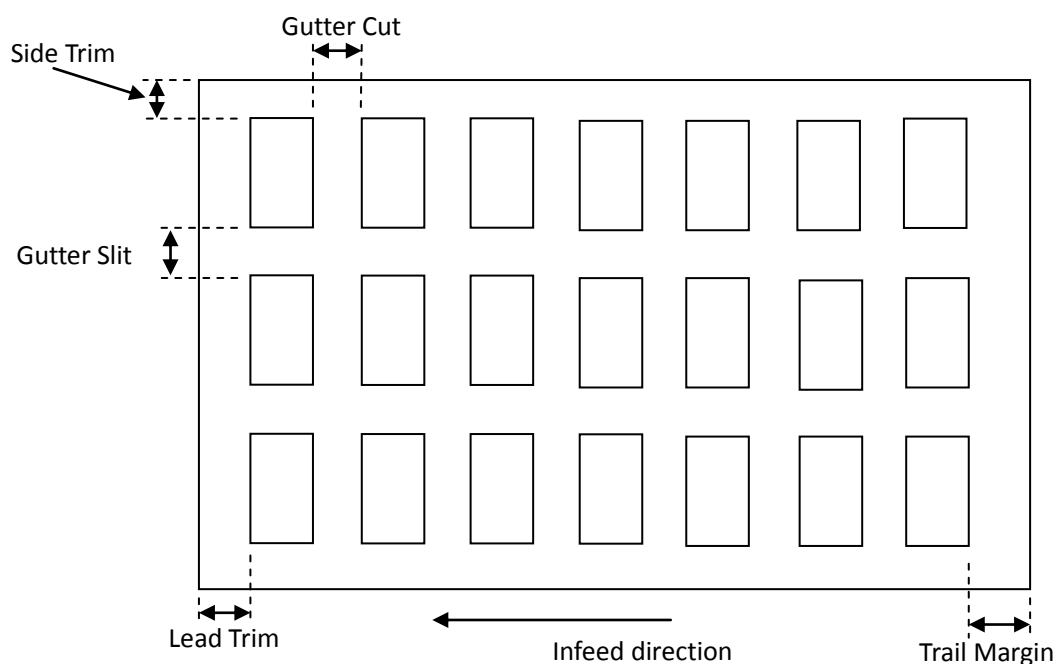
Side Trim: 3.2mm~38.5mm.Usually, we don't recommend our user to set too large side trim. Because the larger trim bleeding will not be able to drop to the waste bin smoothly.

Gutter Cut: 3mm~15mm.It will be automatically chopped up if it was set over 15mm.

Gutter Slit: 0 or 5mm~15mm.

Trail Margin: 5mm~15mm.

***It's very important to keep this parameters in mind so that our user can set up a successful template.**



4. After that, we can set the crease line by using 1st crease bar-tool if needed.

4.Set 1st Crease module

1

0

2

0

3

0

4

0

Back

Next

mm

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent



5. We can also use the 2nd crease bar-tool to do the crease job. It depends. Usually, we can just set the crease job on 1st crease bar-tool.

5.Set 2nd Crease module

1

0

2

0

3

0

4

0

Back

Next

mm

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent



6. Then we can set the linear automatic partial perforator data. PST is the position from the side edge. You can set two partial perf lines on one card. But it depends. If the card is longer, the user can set more lines.

***Please notice that our user only need to input the data for linear automatic partial perforator1. Perforator2 will calculate the data for itself automatically.**

6.Set perforate for Card

1

2

PST :

Back Next

mm			
1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

7. Then we can set the cross partial perforator. First, we need to input the position for the perf line. It's calculated from the card's lead.
In this picture, you can set two cross partial perf lines on one card. But it depends on the card width.

7-1.Set Cross Partial Perf Position for card

1

0

...

2

0

...

Back

Next

mm

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

And then click the , it will enter the following interface. It's used to set the start and end of the partial perf lines.

7-2.Set Cross Partial Perf Position for card

1

0

0

2

0

0

Return

mm

1	2	3	-
4	5	6	+
7	8	9	Esc
.	0	C	Ent

5.3 Limit for tail cut (Trail margin)

In order to increase the cutting accuracy, 8336BSC and 9375BSC has been improved and modified on the machine. it has a lot improvements on cutting accuracy compared with 8335BSC/8330BSC.

However, the modifications will lead to a risk of jam near cutter if our user made templates with a longer tail over 15mm. As for 8330BSC/8335BSC, it will chop up the tail if it was over 15mm. **However, it will NOT chop up the tail on 9375BSC if it was over 15mm.** This is very important.

As shown in the following picture, it will show different results on 8335BSC and 8336BSC/9375BSC for the same setting. **9375BSC will not accept a template with multi-cuts on tail.**

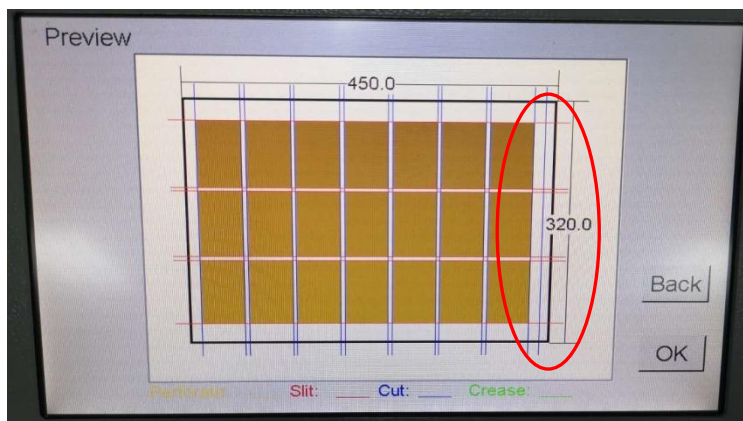


Fig. 9375BSC

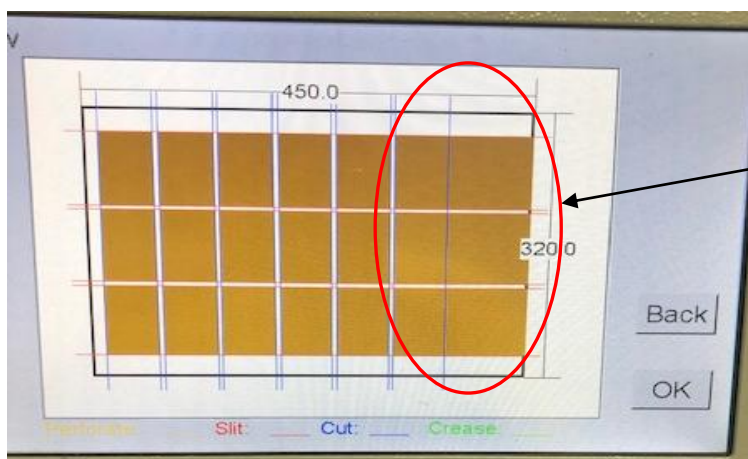


Fig. 9375BSC

9375BSC will not chop up the tail if it was over 15mm.

In order to guide our user operating and making templates correctly, we introduce more details below. It will help to decrease the confusion when making new templates.

Example 1:

Sheet size: 320mm (Width) * 450mm (Length)

Lead trim: 15 mm

Gutter cut: 5mm

Product length: 54mm

We use smart input to make this template, it will be all normal until we input the lead trim and gutter cut . If we set the lead trim 15mm and gutter cut 5mm, it will give the preview template like Fig.4. It looks abnormal.

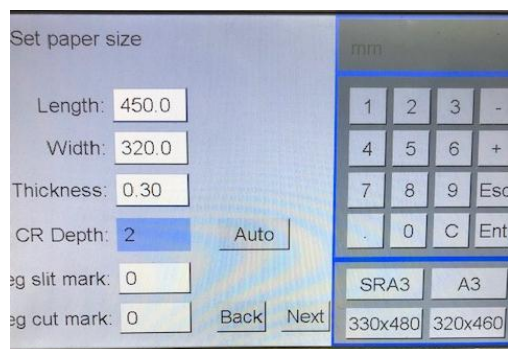


Fig.1

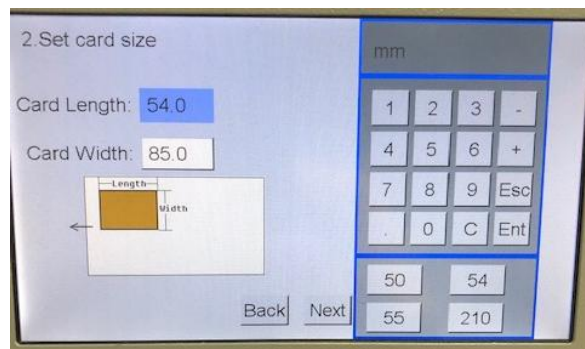


Fig.2

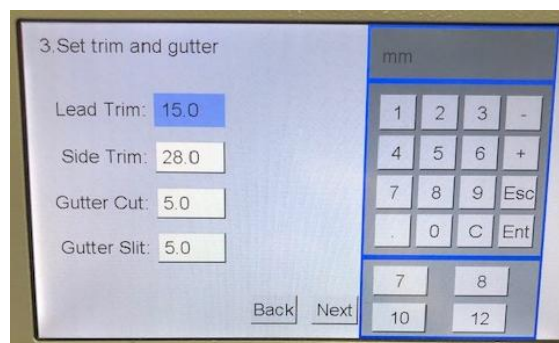


Fig.3

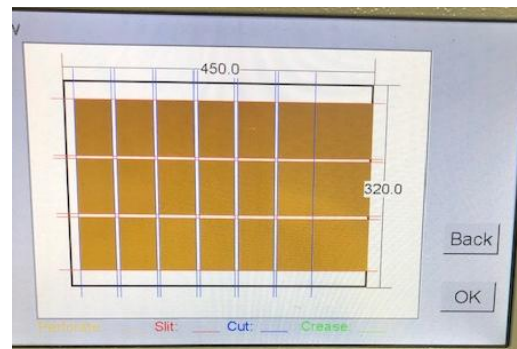
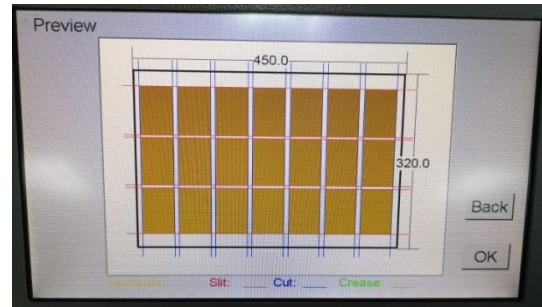


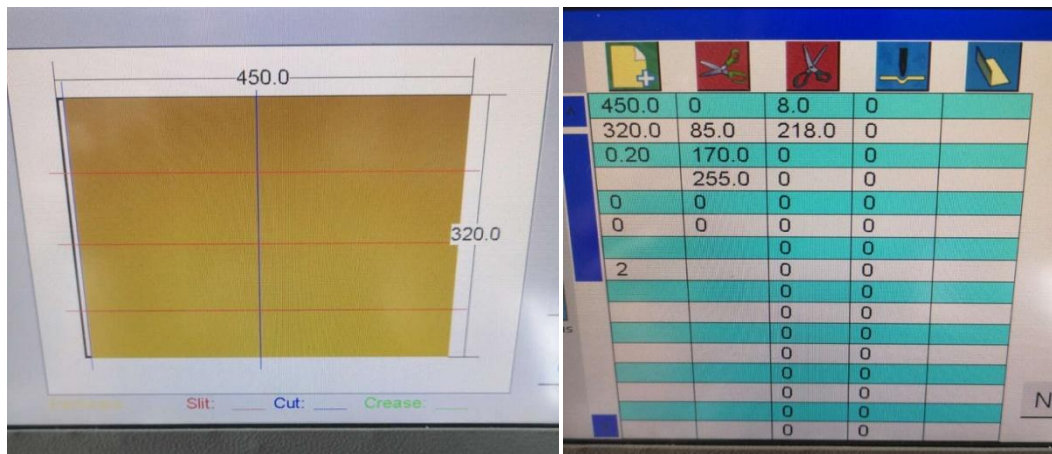
Fig.4

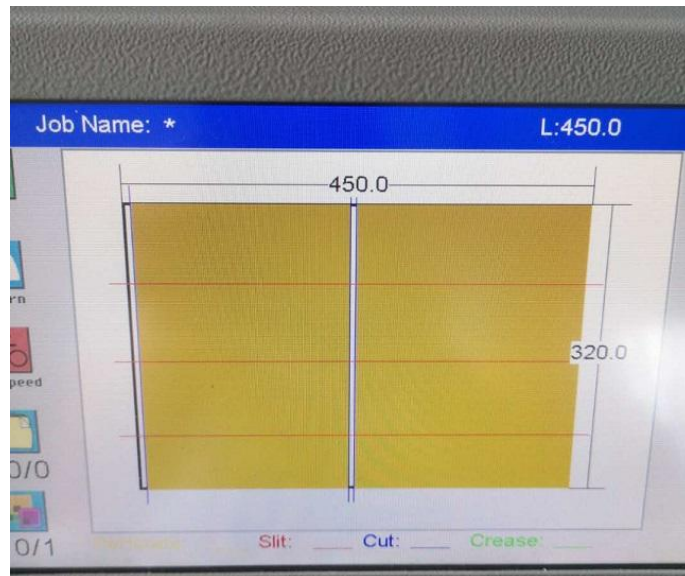
As we can calculate, the tail will left $(450-15-5*6-54*7=27\text{mm})$. It's already over 15mm. So that's why it looks like it won't finish correctly.

To solve this confusion, we can change the values of lead trim or gutter cut. For example, we can set the lead trim 10mm and the gutter cut 8mm .Then the tail will left $(450-10-8*6-54*7=14\text{mm})$.That is to say, we will need to change the lead trim or gutter cut to proper values to make the tail in the scale of 5mm~15mm. As shown in the following, it will become normal after we change the value of lead trim and gutter cut.



Product length: 210mm





5.4 Save or Recall a Template

5.4.1 Save a Template



Press icon to save template to the user template storage.

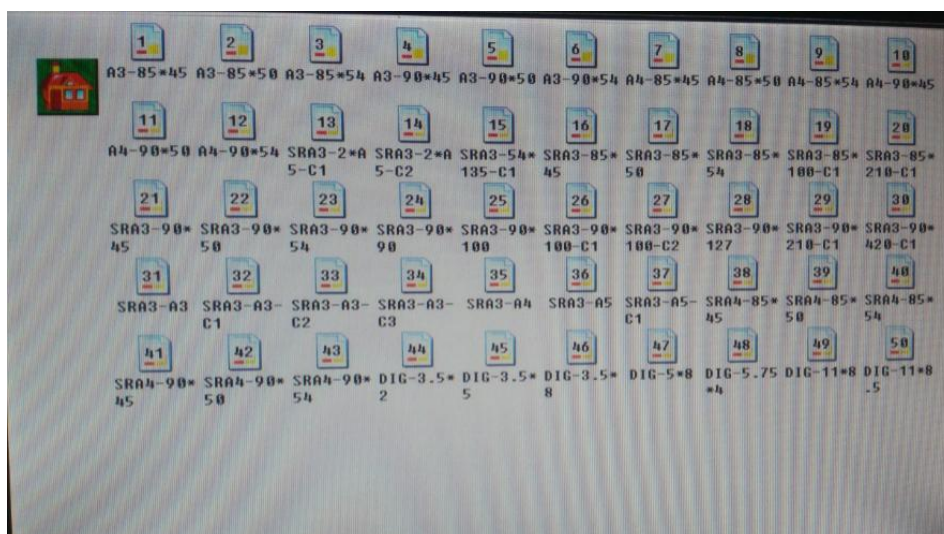
5.4.2 Recall a Template



Press icon to recall a user template from the user template storage.



Press icon to recall a factory template from the preset template storage.



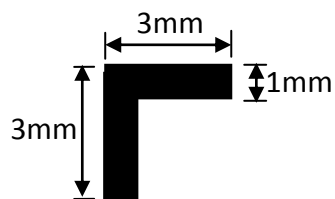
Chapter 6 Important features

6.1 Register mark

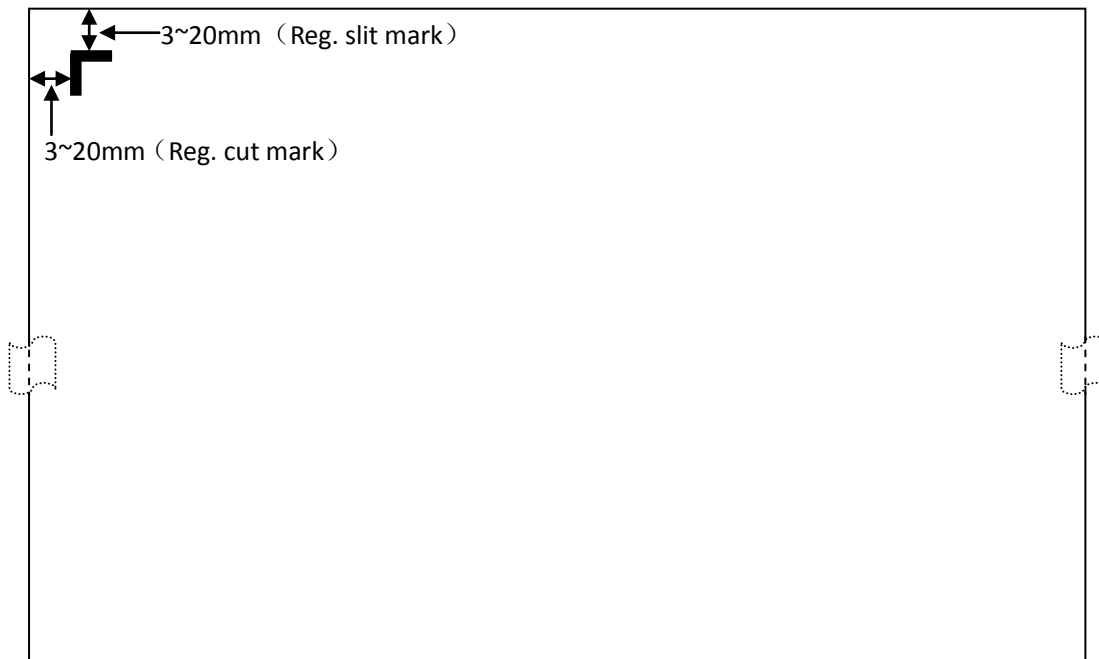
Register mark is used for the compensation of the cutting and slitting accuracy. Because the register mark is printed together with the customer's effective content, they will shrink or stretch together. It will help the cutter and the slitter to cut and slit at the correct position.

6.1.1 Register mark specifications

The size of the register mark is shown in the following picture.

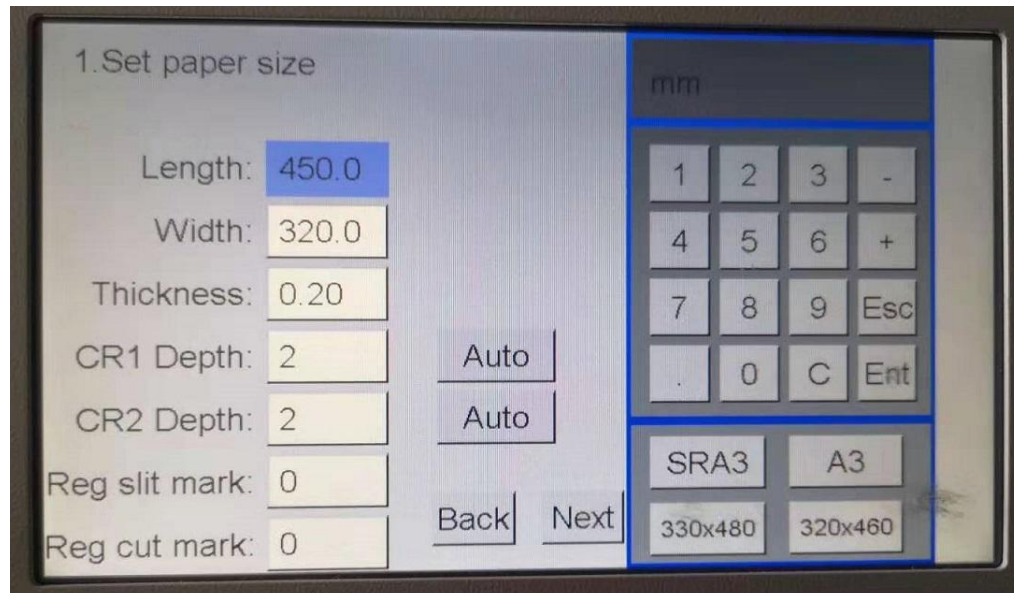


The position of the register mark is shown in the following picture. The distance from both edges is from 3mm~20mm.

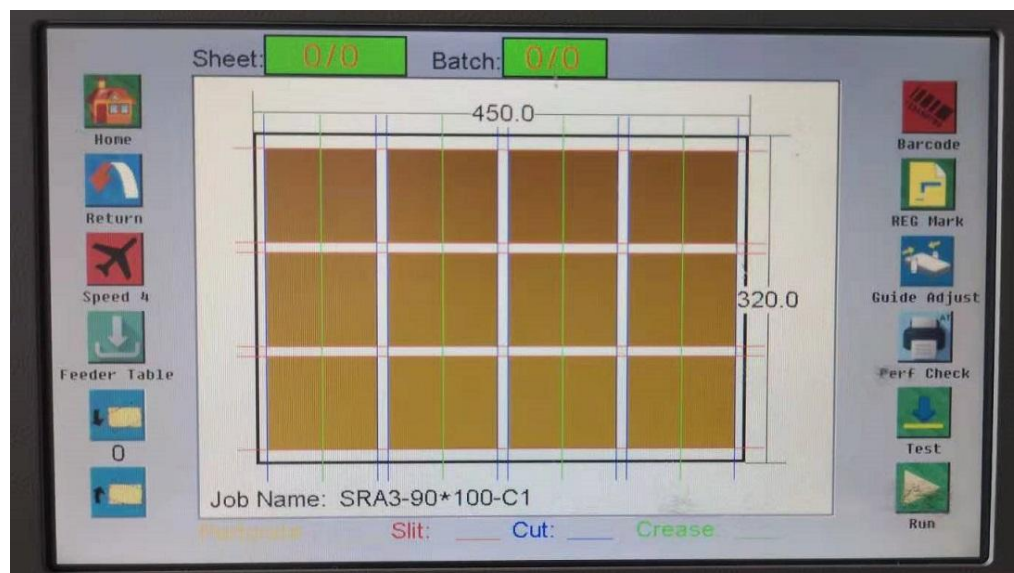


6.1.2 How to activate the register mark

After we have printed the register mark correctly on the sheet, we need to input the Reg. slit mark and the Reg. cut mark values first.



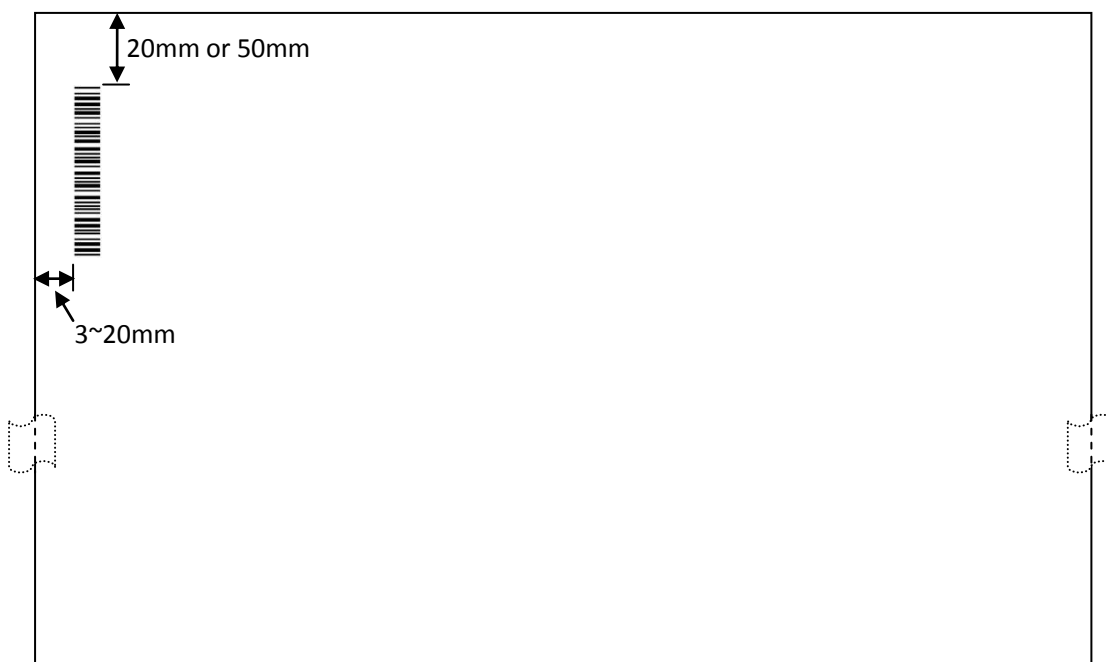
Then go to the standby interface to select the REG Mark icon . If it's activated successfully, the icon will come up a "✓" on it. If not, we will need to check whether we already input the values on Reg. slit mark and Reg. cut mark.



6.2 Barcode Reader

6.2.1 Barcode specifications

As shown in the following picture, the width and the length should be in the certain scale. Please notice that the detector may fail to read the barcode if it's shrunk or stretched.



6.2.2 How to generate the barcode

The barcode is generated by **code39**. It's a kind of standard for barcode.

1) Code 39

Code 39 is one of the protocols that define symbols as barcodes.

The definitions are as following:

Each code consist of 9 bars, 5 in black, 4 in blank.

In 9 bars, only 3 are wide, others are narrow.



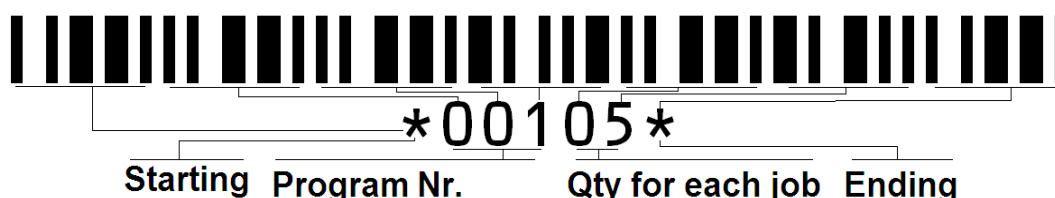
Left drawing is an example of barcode. 9 bars, rightwards, are coded from 1-9, As we can see 5 black bars (Nr. 1 3 5 7 9) , 4 blank bar(Nr. 2 4 6 8)

3 bars (Nr. 2, 5, 7) are wide and other 6 are narrow.

If this bar is read, system will take it as star symbol “*”
 In Dumor barcode, Wide bar is 1mm wide, narrow is 0.5mm wide.

2) Definition of a complete Dumor barcode.

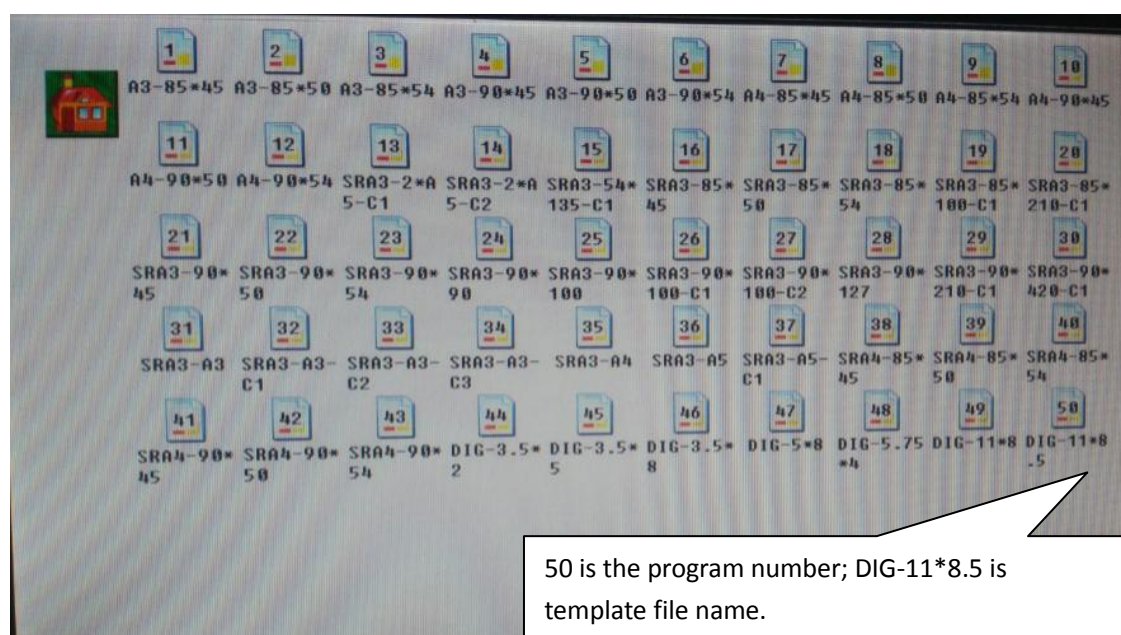
A complete barcode for DUMOR machine consist of 7 symbols:



Symbol 1. Each code is started with a star symbol “*”.

Symbol 2,3,4. Program Number:

The data that controls slitting/cutting/creasing position is saved as files, Factory set program are coded ranging from 1-64, User saved program are coded ranging since 165-228. Once such a number is read, corresponding file will be recalled.



Mind that the code should be triple digit, if it is less than 100, insert a zero(0) before that ,say 065, if it is less than 10, insert 2 zeros(00) before that, say 002.

Symbol 5, 6. Qty for each job

This 2 number is the sheet qty for this job.

Mind that the code should be double digit, if it is less than 10, insert a zero (0) before that, say 06.

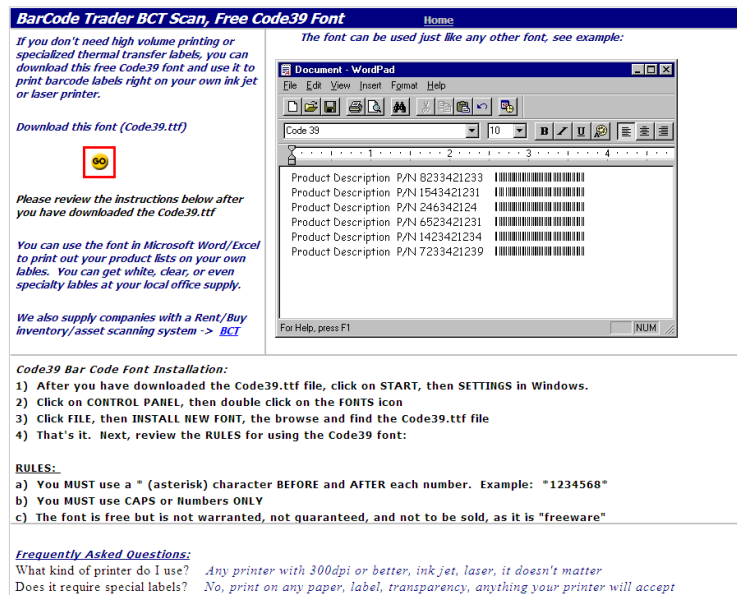
7. Each code is ended with a star symbol “*”.

3) How to create barcodes?

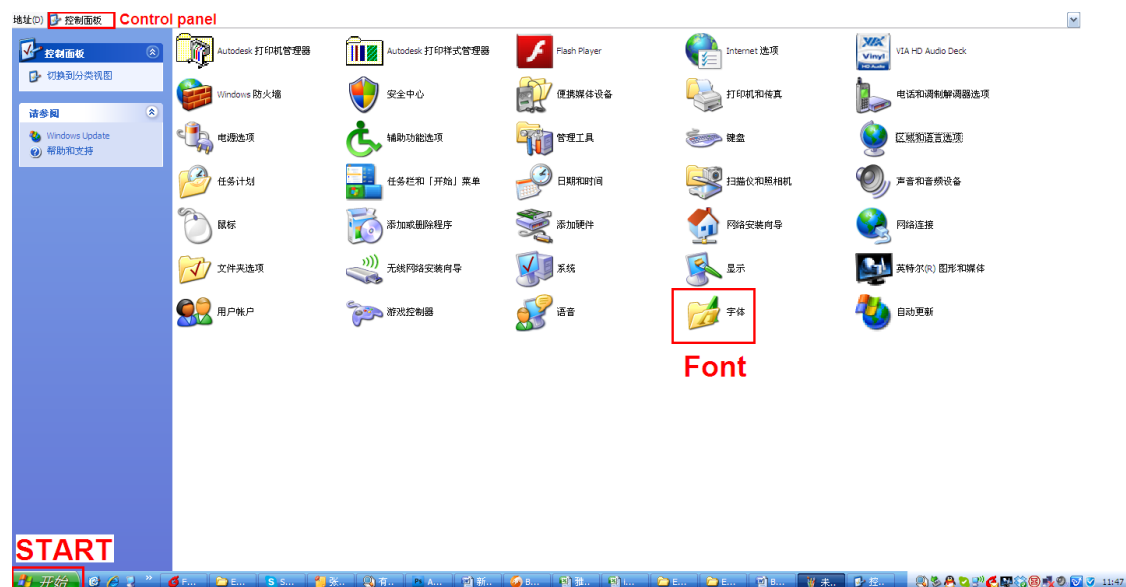
Barcode can be input by keyboard, the point is that user should install a True type Font of code 39.

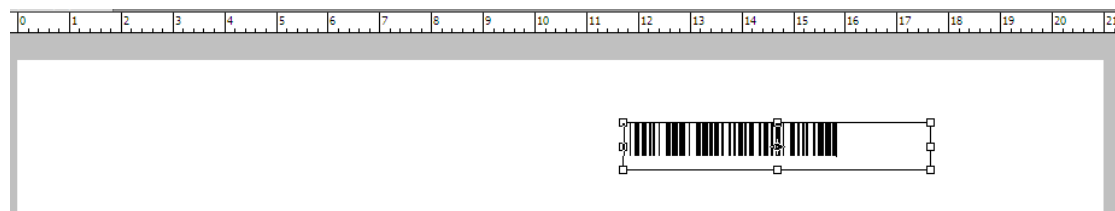
Step1: Load on http://www.barcodetrader.com/bct_scan_barcode.asp

Click the icon in red square, save the file *code39.ttf*



Step 2: Click **START** on windows, enter **CONTROL PANEL**, copy the *code39.ttf* to the folder **Font**. The font will be installed automatically.





4) How to run a job with barcode activated?

- 1) Set up a new template in the machine.
- 2) Create the barcode with the same number.

For example, the template number is 165. Then the barcode should be “*165xx”.

Please notice that the template name is different from the template number.

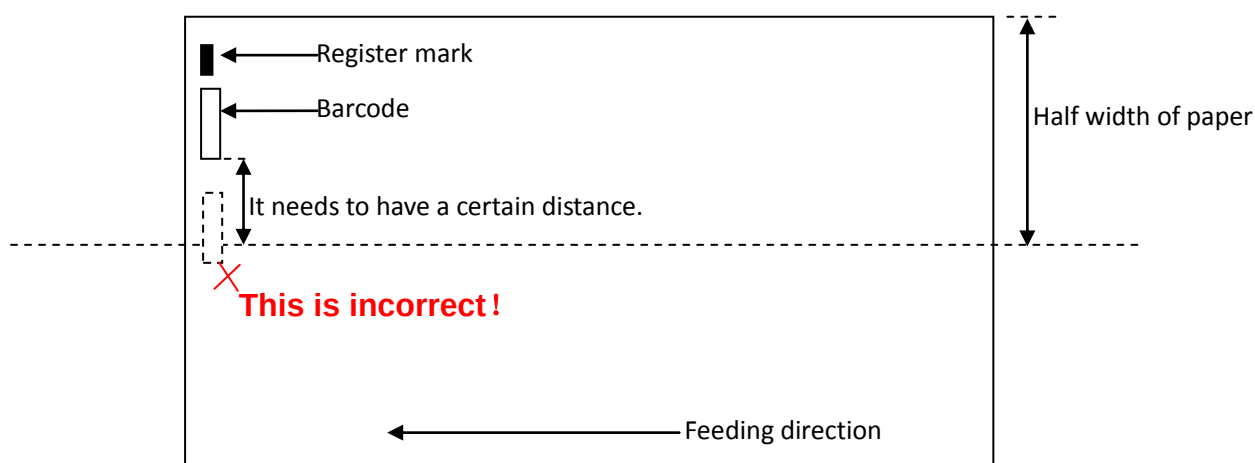
The template number is automatically given by the machine. It can be from 165 to 225. Besides, we can use Trimposer create a barcode job in a much faster way.

If we print the template generated by Trimposer, the barcode or register mark will be automatically generated.

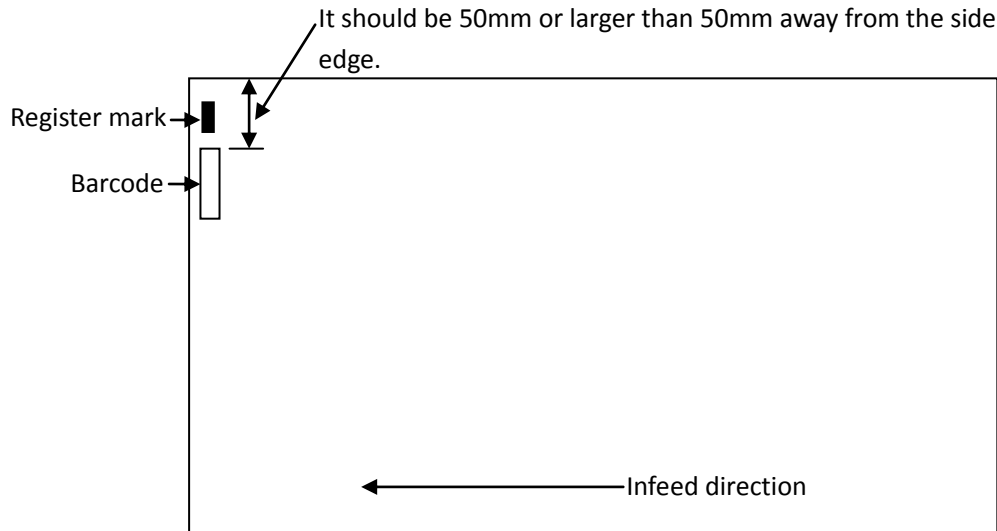
- 3) Then go to the standby interface to click the barcode icon. After the icon show a “v” on it, the barcode has been already activated successfully.

6.3 Important tips for register mark and barcode

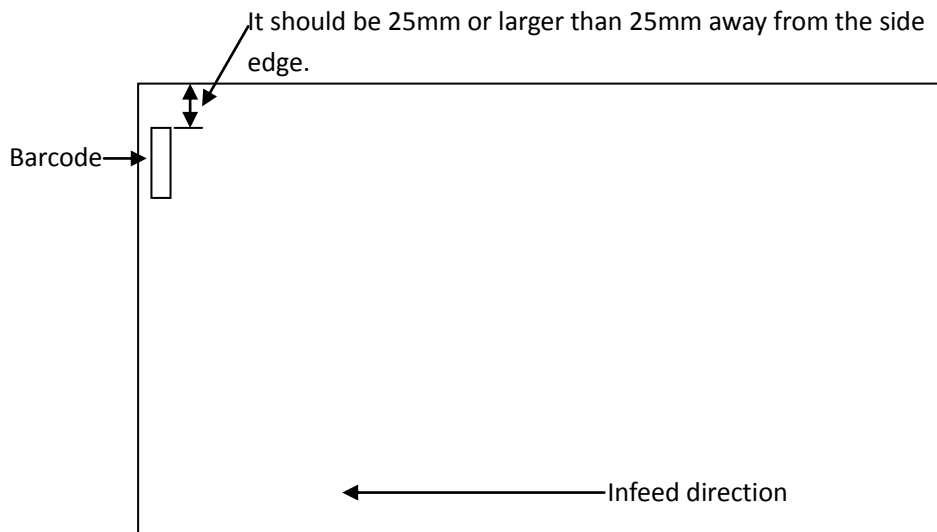
1. The barcode mark cannot be over the half of paper width. For example, If the paper width is 210mm, then the barcode should not be out of the rang of 105mm from the side. If it over the middle line of width, the reader will not be able to detect the barcode.



-
2. When the register mark and barcode are used at the same time, the barcode should be at 50mm or larger than 50mm from the side edge.



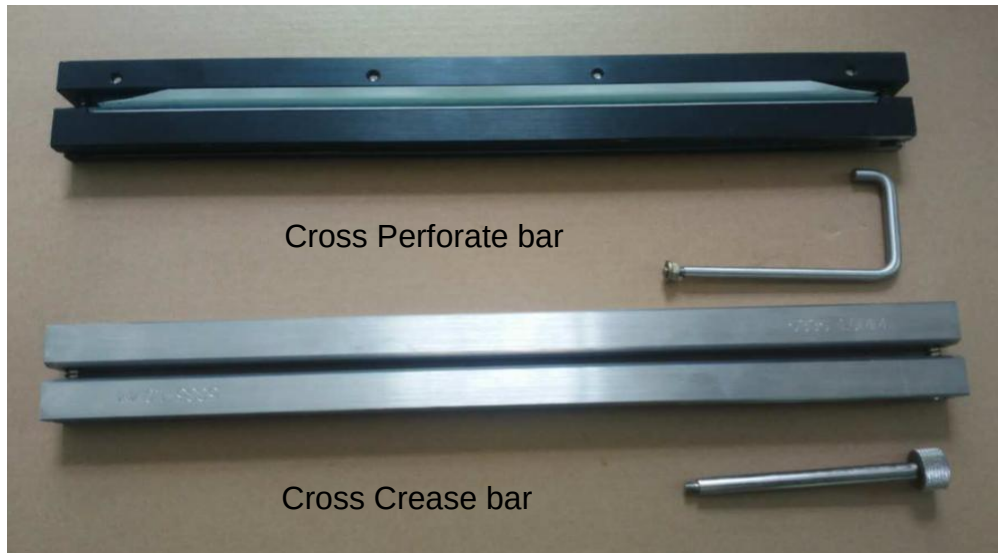
3. When only the barcode is used, it should be at 25mm or larger than 25mm from the side edge.



4. The register mark and barcode reader may fail to read correctly in the following situations:
- 1) the lead of paper curl too much.
 - 2) the register mark and barcode are not printed on white color sheet.
 - 3) There are black spots existing on the register mark and barcode area.
5. When the barcode and register mark are used at the same time, the running speed will drop.

Chapter 7 Option tools

7.1 Cross tools



There are 2 kinds of cross bar, Crease tool and Perforate tool.

To make the sliding easy, we install the screw on the bar.

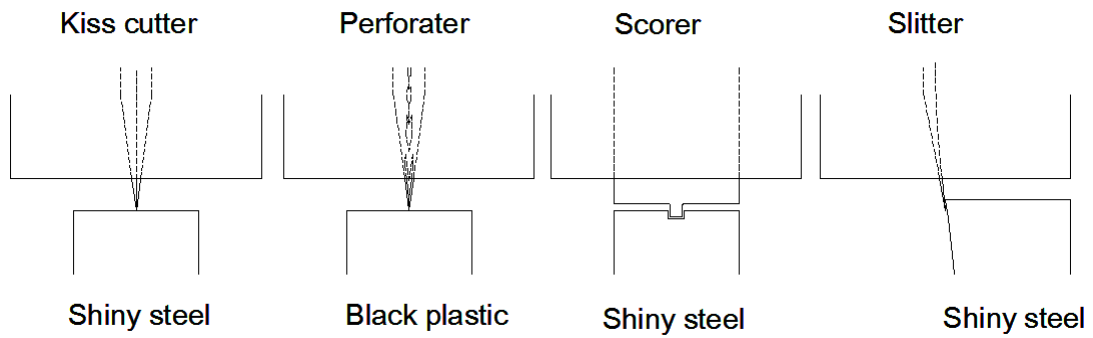
For Crease tools, 1.0mm version is standard, Optional version is 0.6mm, 1.2mm, 1.4mm.

For Perforate tools, there are 8 TPI, 12 TPI and 24TPI version, all are for option.

7.2 Linear Tools

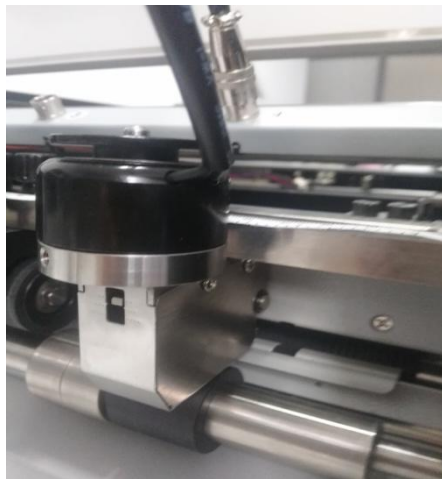


There are 4 kinds of linear optional blades, Scorer, Perforator, Kiss cutter and Slitter. Each tool consists of blade and anvil.



7.3 Strike Perforator

This tool is for the partial perforating. It needs to be moved manually at the cross direction.



7.4 Five rows 55mm business cards assembly

As shown in the following picture, this feature is like a tool box. It's an optional slitting tool for 5 rows 55mm business card.



The function for this tool is in the following chart:

Items	Parameters
Version	V1.00
Part number	370-07-B00
Maximum infeed size (Width*Length)	318mm*500mm
Minimum infeed size (Width*Length)	315mm*210mm
Paper thickness	157g~350g/0.15mm~0.35mm
Paper waviness	Less than 3mm
Maximum output size (Width*Length)	55mm*500mm
Minimum output size (Width*Length)	55mm*48mm
Slit side trim	3.2mm~10mm
Gutter slit	Fixed value: 8mm
Slitting accuracy	±0.3mm (Except very special boards)
The minimum distance between the middle slitters	55mm Fixed distance

Important tips:

1. When the paper width is larger than 318mm, we can use slitter #1 and #6 to cut it to 315mm~318mm.
2. When the paper width is 320mm and the paper thickness is less than or equals to 300g/0.3mm, we don't need to cut it to less than 318mm width.
3. When the paper width is larger or equals to 320mm and the thickness is larger than 300g/0.3mm, we must slit it to 315mm~318mm.

Chapter 8 Trouble Shooting

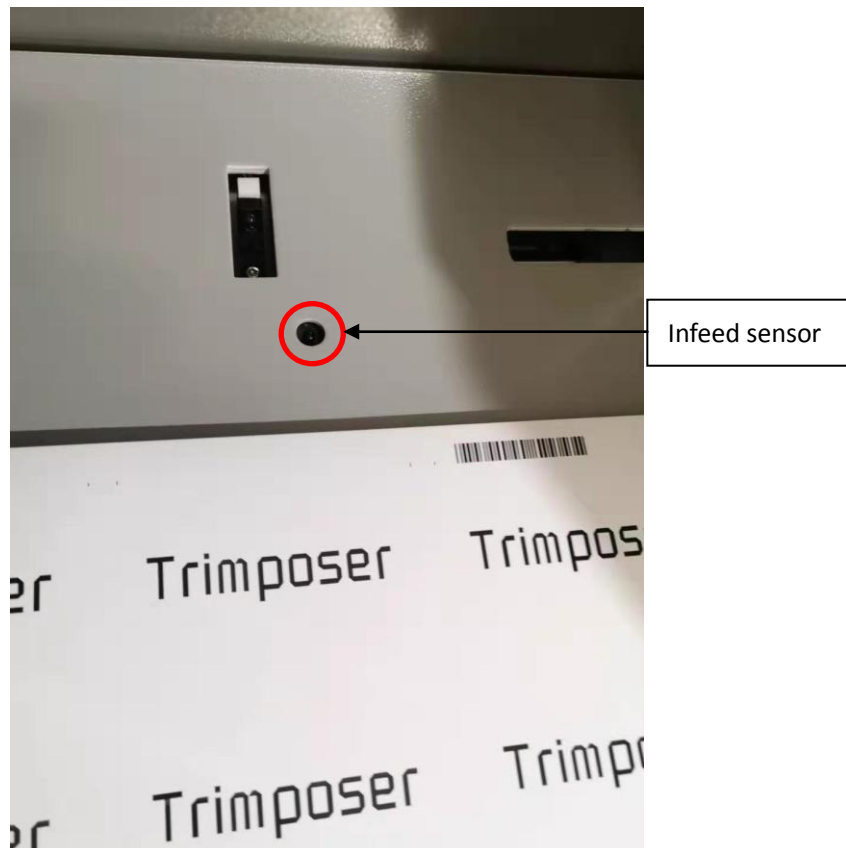
8.1 Show warning messages

1) C-2, Please set paper

It means no sheets are loaded on the feeding table. It shows up after the sheets run out.

When the machine gave this message, we can put some sheets on and then press the RETURN button on the touch screen to clear it.

If there is no paper on but give no warning message, we can try to do some cleanings on the surface of paper detect sensor on feeding table.



2) C-4, Top cover opened at crease unit; C8, Top cover opened at SC unit; C-9, Top cover opened at feeding unit

It means that top cover is opened.

So when this warning message shows up, we need to check whether the top cover was opened or not.

If the top cover was not opened, the trigger on the top cover may not be able to press down the secure switch or the secure switch fail.

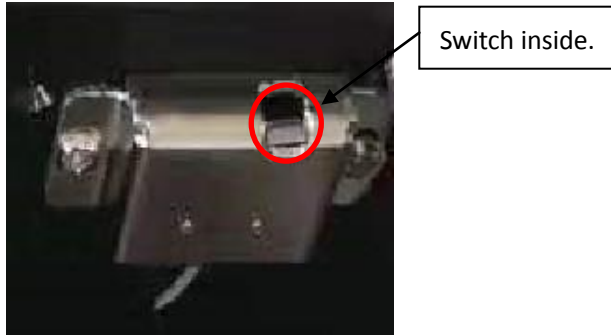


Fig. Secure switch

3) C-5, No bar-tool in station #1 ; C-6, No bar-tool in station #2; C-12, No anvil in position for cross partial perf

It means that there is no crease or perf tool in bar-tool station. We may not inserted any bar-tool in this station or the bar-tool was not in position. There is a secure switch on the other end of station.

When it shows up this message, we need to check whether there is a tool inside or not. If not, put one in. If the bar-tool was already in, we should push the tool to the end until we can head ticking sound.

4) C-7, Side cover opened at crease unit; C-10, Waste bin door opened ; C-11 Tool door opened

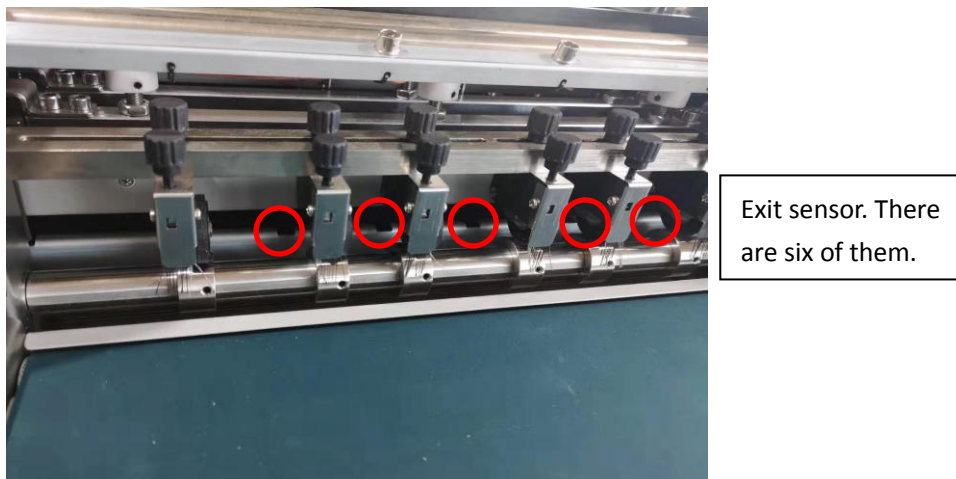
It means that the cover or door is opened.

So when this warning message shows up, we need to check whether the cover or door was opened or not.

If the cover or door was not opened, the trigger on the cover or door may not be able to press down the secure switch or the secure switch fail.

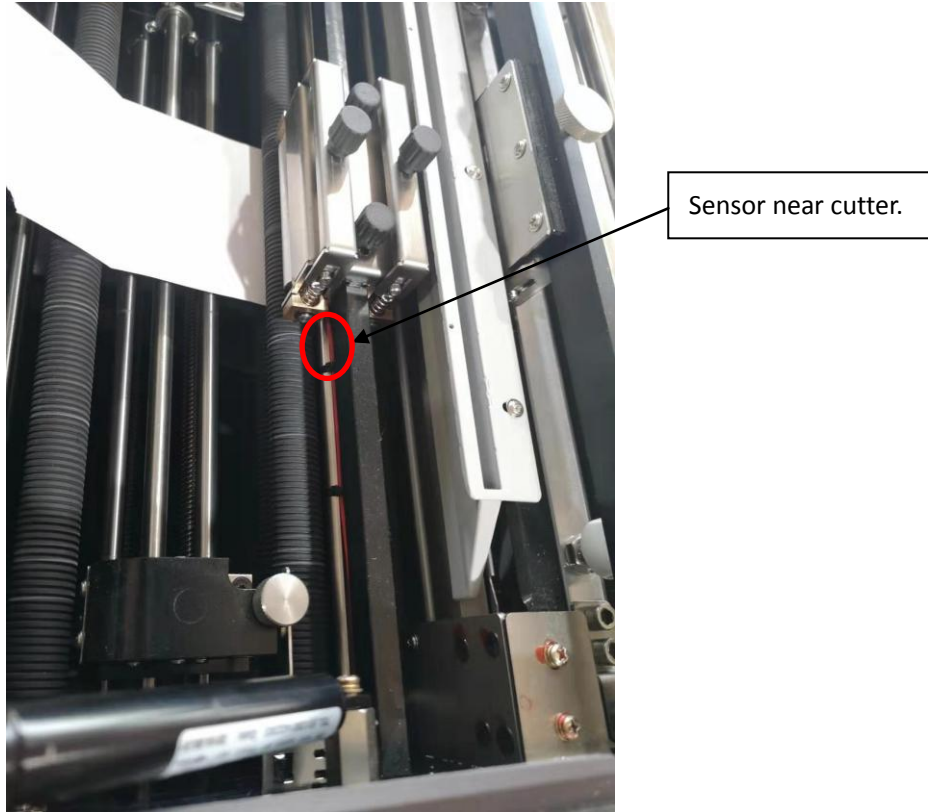
5) C-14, Paper jam near outfeed roller

It means some sheets or dust may be covering in the exit sensor area. Then we can check this area. If there was no paper jam in the area, it may have dust covering the exit sensor. In this case, we can use a air gun or something else to clean the sensor surface.



6) C-15, Paper jam near cutter

It means some sheets or dust may be covering in the sensor near the cutter area. Then we can check this area. If there was no paper jam in the area, it may have dust covering the sensor. In this case, we can use a air gun or something else to clean the sensor surface.

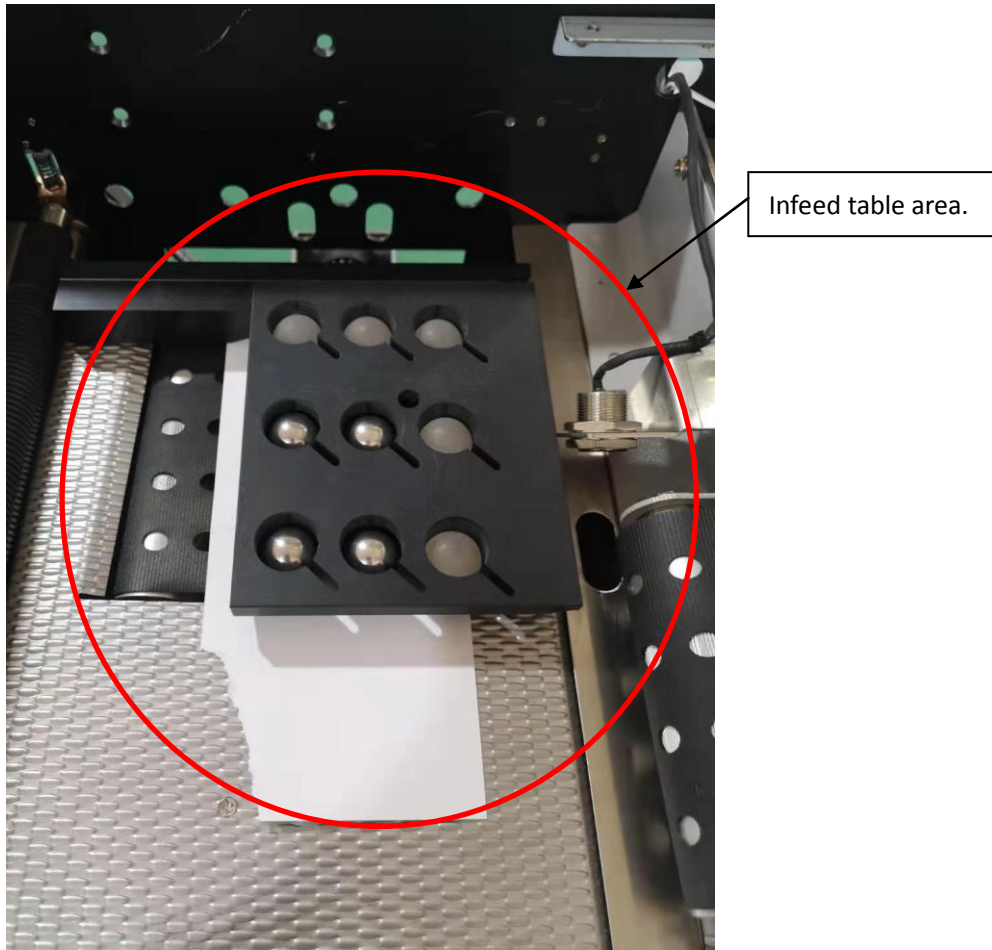


7) C-16, Double feed jam

It means double feed. If there is no double feed, the double detect sensor may need to recalibrate. If it cannot be fixed, we can enter the user mode to switch off the double feed detector.

8) C-17, Paper jam at infeed table

It means some sheets or dust may be covering in the infeed sensor area. Then we can check this area. If there was no paper jam in the area, it may have dust covering the infeed sensor. In this case, we can use a air gun or something else to clean the sensor surface.



9) E-1 , Elevator motor error

When it happened, we can try turning off the machine and then turning on first. If it didn't work, try the following steps:

If it happened when the feed table stop on the top, we can press the green release button and the downward button at the same time to make the table free from the top.

If it happened when the feed table stop at the bottom, we can press the green release button and the START button at the same time to make the table free from the bottom.

10) E-3, Crease motor error for station #1; E-4, Crease motor error for station #2

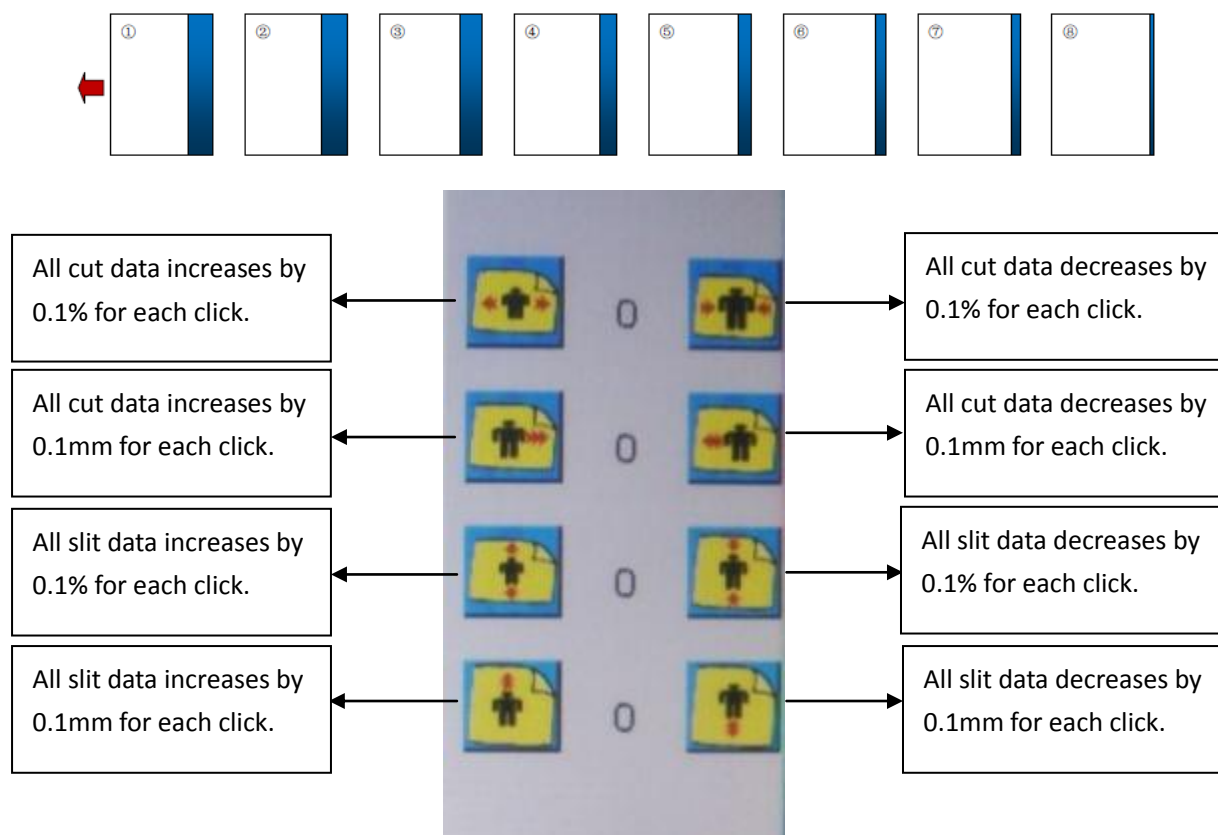
This is probably caused by the thickness of paper and the crease depth. As for thicker paper, it may show this error if we set a very large crease depth. In this case, we can try a smaller crease depth.

11) E-5, Cutter motor error

This is probably caused by thicker paper or special material. If the thickness of paper is over the parameter or we run some very hard material, the cutter is unable to cut it off. And then it will show this error.

8.2 Images Shrink/Stretch or Shift Caused by Printing

The cards in same column are like following picture:



Steps: Assume the first card is the standard card. The later card's color edge is smaller and smaller.

1. Open the register mark function if the register mark has been printed onto the sheet.



2. Click the icon  to increase the cut data by 0.1% each time and run one sheet to check until the color edge is nearly the same.

3. If the color edge is nearly the same, but the color edge is bigger than standard



card. Click the icon  to decrease cut data by 0.1mm each time and run one sheet to check until it reaches correct size.

Chapter 9 Maintenance

Maintenance is important for machine. Essential maintenance is in favor of use life and efficiency of machine. It consists of user maintenance and technician maintenance.

User maintenance

Clean the roller with wet cloth and infeed, outfeed sensors with air gun while machine used for long time.

Meanwhile the working environment should be clean and dry, as well as no light shining directly.

Technician maintenance

Do the technician maintenance regularly is also very important. It mainly finished by technician, including checking the wear and tear parts, replace the broken parts, adding lubricant and so on.

More details referred to technician manual.